



Digital-logic-capsule sag

Communication Techniques (Pokhara University)



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Pokhara University
Faculty of science and technology

Subject . Digital Logic Capsule
Computer Engineering I Semester
Software Engineering I Semester
Ele and Eletrical Engineering III Semester

Name..

Subject..

Semester..

By. Er. Manoj Kumar Singh
Availeval. Bijayapur library

Design a combinational ckt which takes three i/p numbers and produces an output equal to square of the input. [2013 Fall]

Analysis:

Max No. = 111 = 7 square of 7 = $7^2 = 49$

49 in binary = 110001

Total no. of bits in o/p seq = 6.

	i/p			o/p						
	A ₂	A ₁	A ₀	S ₅	S ₄	S ₃	S ₂	S ₁	S ₀	
0	0	0	0	0	0	0	0	0	0	square of 0 = 0
1	0	0	1	0	0	0	0	0	1	square of 1 = 1
2	0	1	0	0	0	0	1	0	0	square of 2 = 4
3	0	1	1	0	0	1	0	0	1	square of 3 = 9
4	1	0	0	0	1	0	0	0	0	square of 4 = 16
5	1	0	1	0	1	1	0	0	1	square of 5 = 25
6	1	1	0	1	0	0	1	0	0	square of 6 = 36
7	1	1	1	1	1	0	0	0	1	square of 7 = 49

Truth Table

Now let us draw the K-Map for S₀ for S₂

A ₂ \ A ₁ A ₀	00	01	11	10
0		1	1	
1		1	1	

$$S_0 = A_0$$

A ₂ \ A ₁ A ₀	00	01	11	10
0				1
1				1

$$S_2 = A_1 \bar{A}_0$$

for S₃

A ₂ \ A ₁ A ₀	00	01	11	10
0			1	
1		1		

$$S_3 = A_0 A_1 \bar{A}_2 + A_0 \bar{A}_1 A_2$$

for S₄

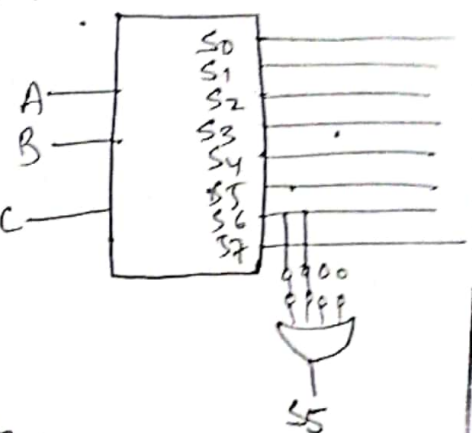
A ₂ \ A ₁ A ₀	00	01	11	10
0				
1	1	1	1	

$$S_4 = \bar{A}_1 A_2 + A_0 A_2$$

for S₅

A ₂ \ A ₁ A ₀	00	01	11	10
0				
1			1	1

$$S_5 = A_1 A_2$$



Now, let us draw the logic diagram.

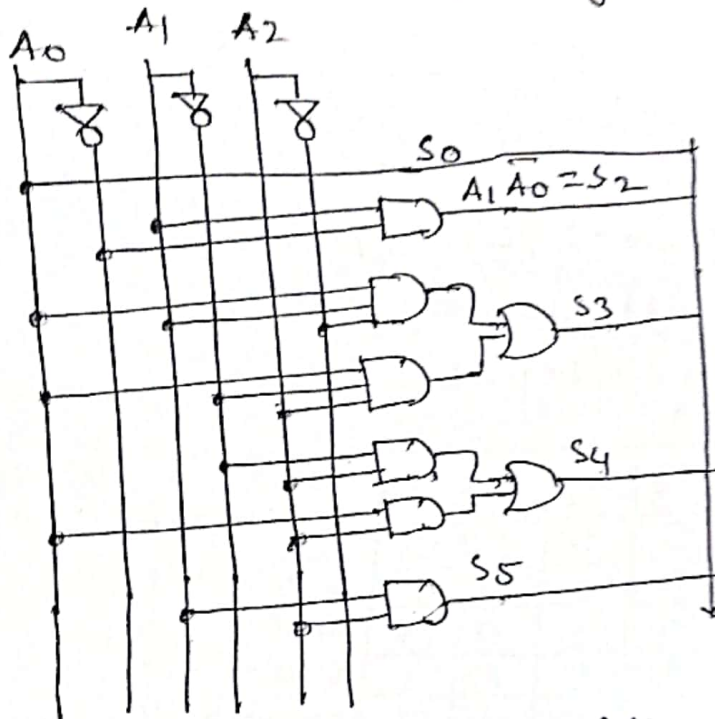


Fig:- logic diagram of the given 3 bit square wave o/p.

2. Design a Combinational logic CKT with 3 inputs that provide 1 o/p when exactly two variables are 1.

Let us draw the Truth Table.

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

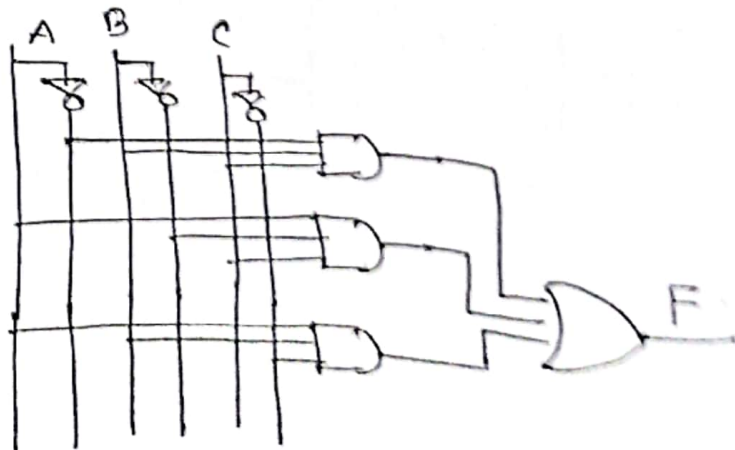
Truth Table.

Now K-Map for F

BC	00	01	11	10
A=0			1	
A=1		1		1

$$F = \bar{A}BC + A\bar{B}C + ABC$$

Now draw the logic diagram.



Q. Design a combinational ckt with three inputs x, y, z and three outputs A, B, C when the binary input is 0, 1, 2 or 3, the binary output is two greater than the input. when the binary input is 4, 5, 6 or 7, the binary output is two less than the input.

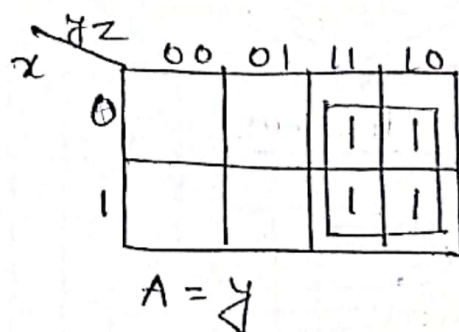
Soln.

	x	y	z	A	B	C
0	0	0	0	0	1	0
1	0	0	1	0	1	1
2	0	1	0	1	0	0
3	0	1	1	1	0	1
4	1	0	0	0	1	0
5	1	0	1	0	1	1
6	1	1	0	1	0	0
7	1	1	1	1	0	1

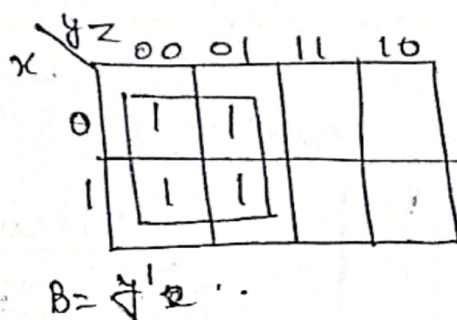
Truth Table

Note
 Acc. to ques
 0, 1, 2 or 3
 Add 2 to
 4, 5, 6 or 7
 Subtract 2
 4-2, 5-2
 6-2

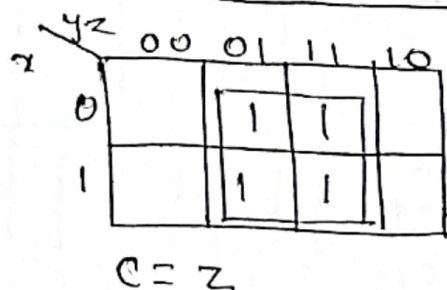
Now, K Map for Output A



K-Map for Output B



Again, K-Map for Output C



Now draw the logic diagram

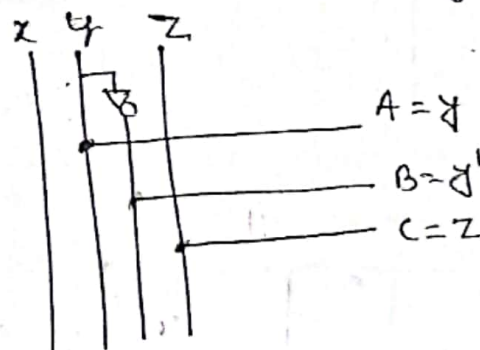


Fig. logic diagram of given comb. ckt

Design a combinational ckt with three inputs and one output.

- The output is 1 when binary value of the inputs is less than or equal to 3. The output is 0 otherwise.
- The output is 1 when the binary value of the input is an odd number.
- The output is 1 when the binary value of the input is an even number.

Solution: → (a)

x	y	z	F
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

Truth table.

Now, write the K-map for F

x \ z	00	01	11	10
0	1	1		1
1				

$$F = x'y' + x'z'$$

Now, Draw the logic diagram of the Boolean function.

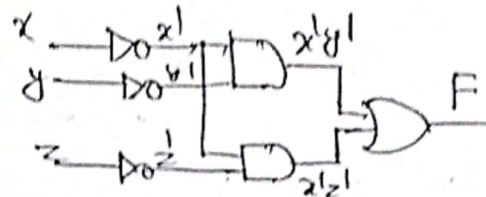


Fig: logic diagram

(b)

for odd No. O/P = 1

x	y	z	F
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1

K-Map for O/P F

x \ z	00	01	11	10
0	0	1	1	0
1		1	1	

$$F = z$$

Now logic ckt

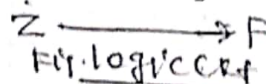


Fig: logic ckt

(c)

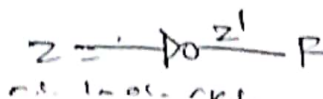
for even No. O/P = 1

x	y	z	F
0	0	0	1
1	0	0	0
2	0	1	0
3	0	1	1
4	1	0	1
5	1	0	0
6	1	1	1
7	1	1	0

K-Map for O/P F

x \ z	00	01	11	10
0	1			1
1	1			1

$$F = z'$$



$F(w,x,y,z) = \sum(1,6,7,8,11,13)$ with don't care condition
 $d(w,x,y,z) = \sum(0,2,3,4,10,12)$ then implement the function using NOR gates only. [2012 Fall]

Solⁿ: Here given:

$$F(w,x,y,z) = \sum(1,6,7,8,11,13)$$

$$d(w,x,y,z) = \sum(0,2,3,4,10,12)$$

YZ	(y+z)	(y+z̄)	(ȳ+z)	(ȳ+z̄)
00	00	01	11	10
01	00	01	11	10
11	00	01	11	10
10	00	01	11	10

WX
 (w+x) 00
 (w+x̄) 01
 (w̄+x) 11
 (w̄+x̄) 10

(I group)
 (II group)
 (IV group)

$$F = (y+z) \cdot (\bar{w}+\bar{x}+y) \cdot (w+\bar{y}) \cdot (x+\bar{y}) \cdot (w+x)$$

Now Double complementing on both side we get

$$\begin{aligned} \overline{\overline{F}} &= \overline{(y+z) \cdot (\bar{w}+\bar{x}+y) \cdot (w+\bar{y}) \cdot (x+\bar{y}) \cdot (w+x)} \\ &= \overline{(y+z)} + \overline{(\bar{w}+\bar{x}+y)} + \overline{(w+\bar{y})} + \overline{(x+\bar{y})} + \overline{(w+x)} \end{aligned}$$

Now, Implementing the function using NOR gate only.

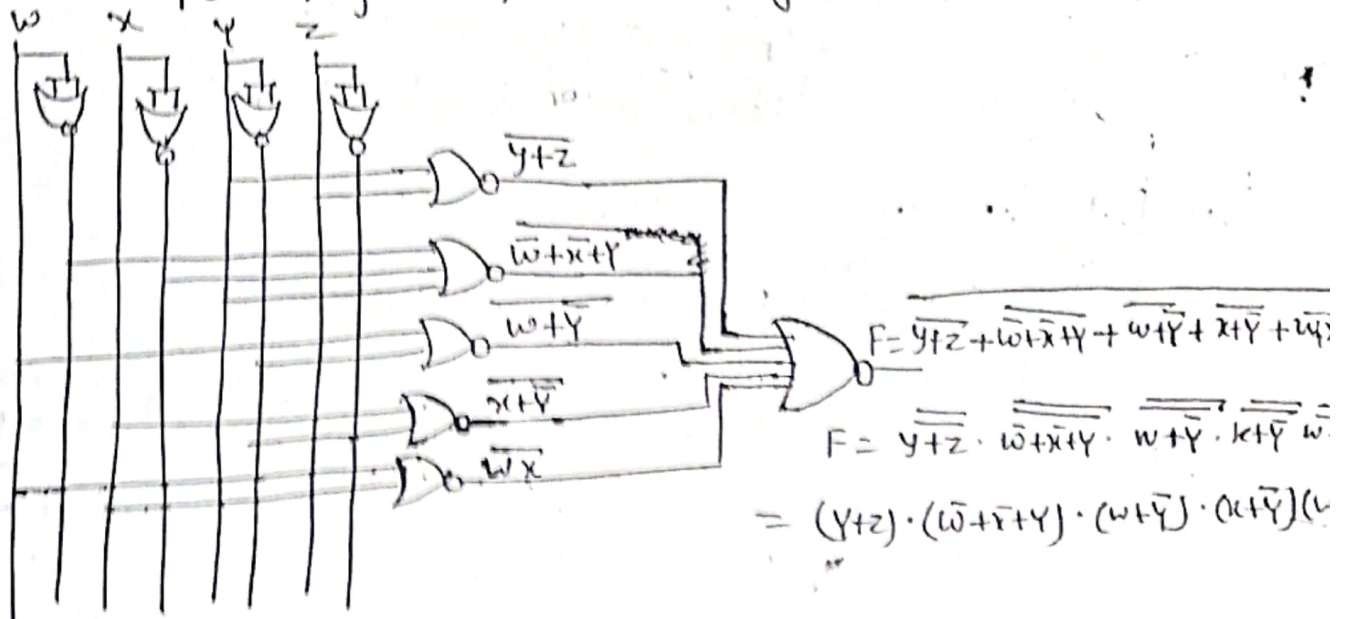


Fig. logic diagram using NOR gate only.

Prepared by: U. K. Singh

* Use K-Map to simplify the given function. Implement the simplified function using NOR gate only.

$$F(A, B, C, D) = \sum(1, 3, 8, 9, 12, 13, 14, 15) \text{ \& Don't care}$$

$$d(A, B, C, D) = \sum(2, 7, 10) \text{ (2012 Spring)}$$

Solⁿ → Here, Given

$$F(A, B, C, D) = \sum(1, 3, 8, 9, 12, 13, 14, 15)$$

$$d(A, B, C, D) = \sum(2, 7, 10)$$

AB \ CD	00	01	11	10
00		0	0	X
01			X	
11	0		0	0
10	0	0		X

$$F = (B + C + \bar{D}) \cdot (A + \bar{C} + \bar{D}) \cdot (\bar{A} + \bar{B} + \bar{C}) \cdot (\bar{A} + D)$$

Now, Implementing this function by using NOR gate only

Now, Double complementing on both side we get

$$\bar{\bar{F}} = \overline{(B + C + \bar{D}) \cdot (A + \bar{C} + \bar{D}) \cdot (\bar{A} + \bar{B} + \bar{C}) \cdot (\bar{A} + D)}$$

$$= \overline{(B + C + \bar{D}) + (A + \bar{C} + \bar{D}) + (\bar{A} + \bar{B} + \bar{C}) + (\bar{A} + D)}$$

Now, Implementing this function by using NOR gate only

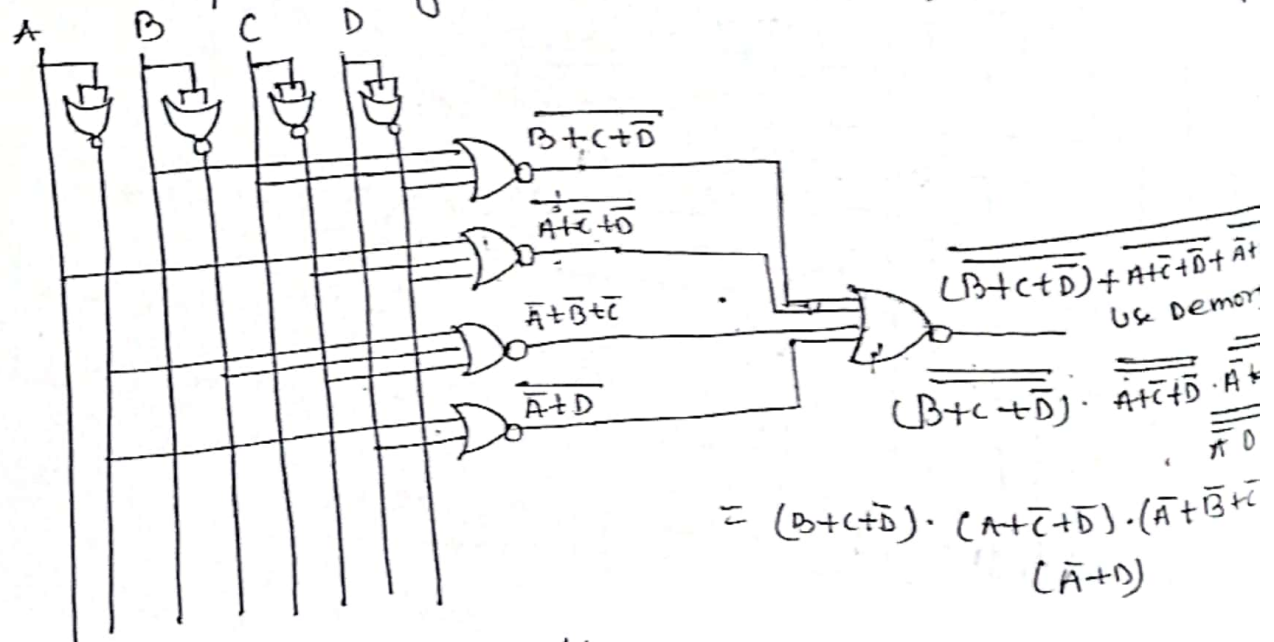


Fig: Logic diagram using NOR gates only.

Prepared By:- En. Manoj Kr. Singh.

① Design a logic function $F(A, B, C, D)$ to implement the Boolean function.

$$F(A, B, C, D) = \sum(1, 3, 7, 11, 15)$$

$$D(A, B, C, D) = \sum(0, 2, 5) \text{ in the term of}$$

(i) SOP

(ii) implement with NAND-NAND gate [2013 Fall] Only.

Soln: →

$$F(A, B, C, D) = \sum(1, 3, 7, 11, 15)$$

$$D(A, B, C, D) = \sum(0, 2, 5)$$

	CD	00	01	11	10
AB	00	X	1	1	X
	01		X	1	
	11			1	
	10			1	

I Group (circled around the 1s in the 11 column)

II Group (circled around the 1s in the 01 column)

$$F = \bar{A}D + CD$$

Now, Double complementing on both side we get

$$\begin{aligned} \bar{\bar{F}} &= \bar{\bar{A}D + CD} \\ &= \bar{\bar{A}D} \cdot \bar{CD} \quad (\text{Using De Morgan's theorem}) \end{aligned}$$

Now, Implement by using NAND gate only.

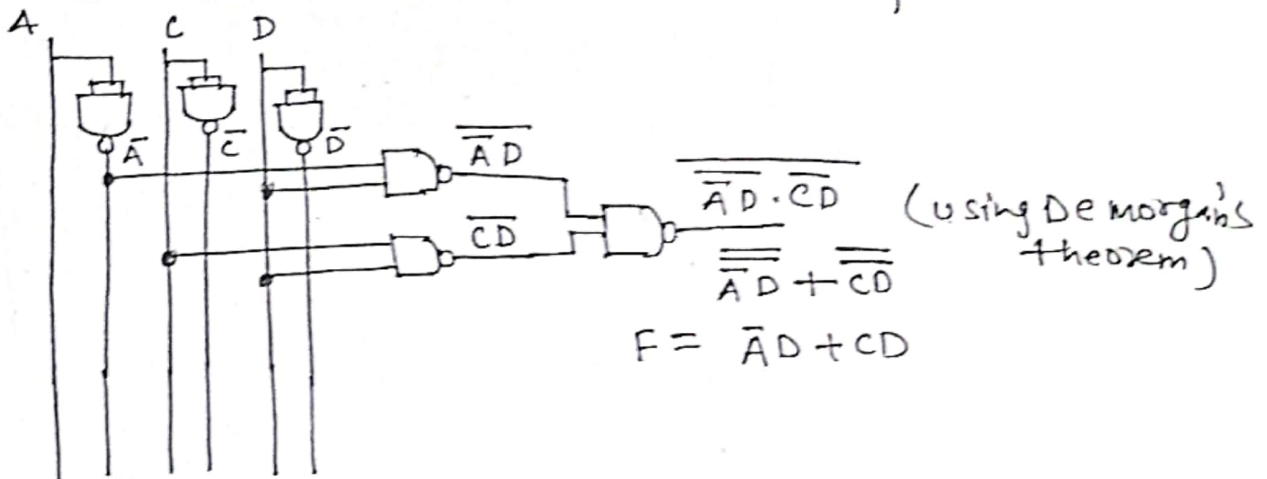


Fig: Logic diagram using NAND gate only

Prepared By: ER. Manoj Kumar Singh

② Obtain a reduced expression for the following $F(A, B, C, D) = \sum m(0, 1, 2, 5, 7, 8, 9, 10, 13, 15)$ and realize the minimized function using NOR gate only.

Soln → Here,

$$F(A, B, C, D) = \sum m(0, 1, 2, 5, 7, 8, 9, 10, 13, 15)$$

CD	(c+d)	(c+d̄)	(c̄+d)	(c̄+d̄)
	00	01	11	10
AB				
00			0	
01	0			0
11	0		0	0
10			0	

$$F = (B + \bar{C} + \bar{D}) \cdot (\bar{B} + D)$$

Now, Double complementing on both sides we get

$$\begin{aligned} \bar{\bar{F}} &= \overline{(B + \bar{C} + \bar{D}) \cdot (\bar{B} + D)} \\ &= \overline{(B + \bar{C} + \bar{D})} + \overline{(\bar{B} + D)} \end{aligned}$$

Now, Implement by using NOR gate only.

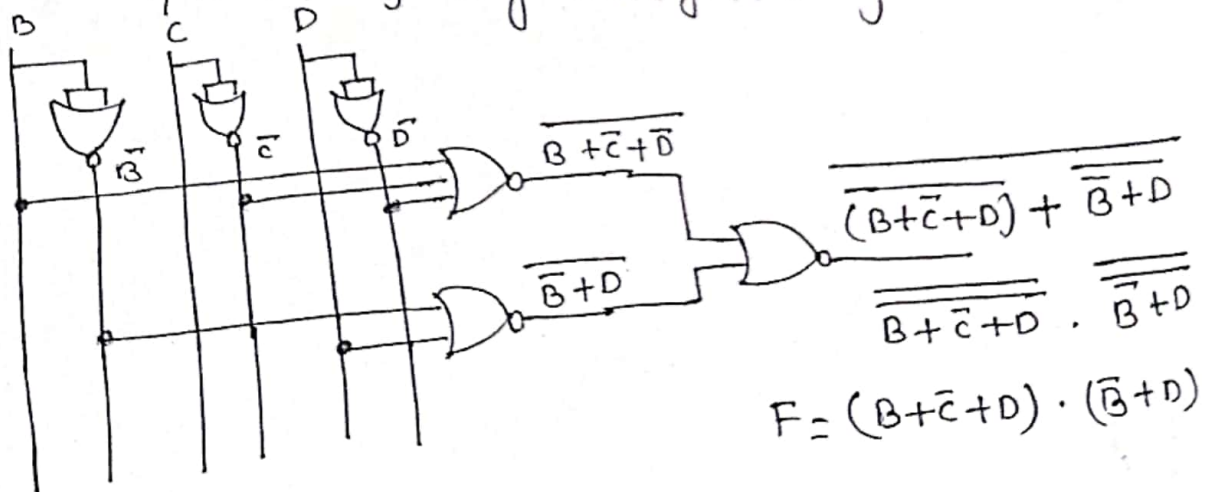


Fig :- logic diagram

(3) Obtain a realization for the following $F(A, B, C, D) = \sum m(0, 1, 2, 8, 10, 11, 14, 15)$ and realize the minimized function using NOR gate only. 3

⇒ Here,

$$F(A, B, C, D) = \sum m(0, 1, 2, 8, 10, 11, 14, 15) \text{ for Maxterm}$$

$$F(A, B, C, D) = \prod M(3, 4, 5, 6, 7, 9, 12, 13)$$

CD	($\bar{C} + \bar{D}$)	($\bar{C} + D$)	($C + \bar{D}$)	($C + D$)
AB	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	0	0	0	0
10	0	0	0	0

$$F = (A + \bar{C} + \bar{D}) (\bar{A} + C + \bar{D}) \cdot (A + \bar{B}) \cdot (\bar{B} + C)$$

Now, Double complementing on both sides we get.

$$\begin{aligned} \overline{\overline{F}} &= \overline{(A + \bar{C} + \bar{D}) \cdot (\bar{A} + C + \bar{D}) \cdot (A + \bar{B}) \cdot (\bar{B} + C)} \\ &= \overline{(A + \bar{C} + \bar{D})} + \overline{(\bar{A} + C + \bar{D})} + \overline{(A + \bar{B})} + \overline{(\bar{B} + C)} \end{aligned}$$

Now, Implement by using NOR gate only.

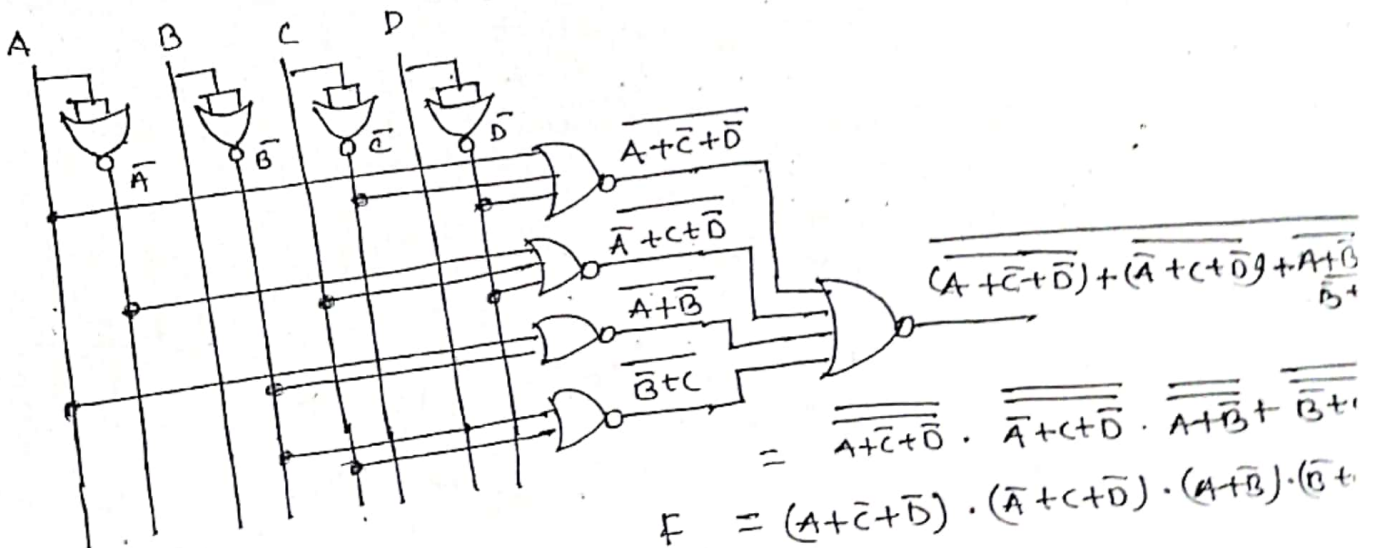


Fig. logic diagram.

Prepared By: ER. Manoj Kumar Singh

3

(4) Design a logic circuit to implement the Boolean function.

$$F(A, B, C, D) = \sum(1, 3, 4, 5, 7, 9, 13, 14, 15)$$

$$D(A, B, C, D) = \sum(0, 2, 8)$$

(i) Sum of product

(ii) Implement with NAND-NAND gate only.

Sol $\rightarrow$ Here, Given

$$F(A, B, C, D) = \sum(1, 3, 4, 5, 7, 9, 13, 14, 15)$$

$$D(A, B, C, D) = \sum(0, 2, 8)$$

CD \ AB	00	01	11	10
00	X	1	1	X
01	1	1	1	
11		1	1	1
10	X	1		

$$F = ABC + \bar{A}\bar{C} + \bar{A}D + \bar{C}D$$

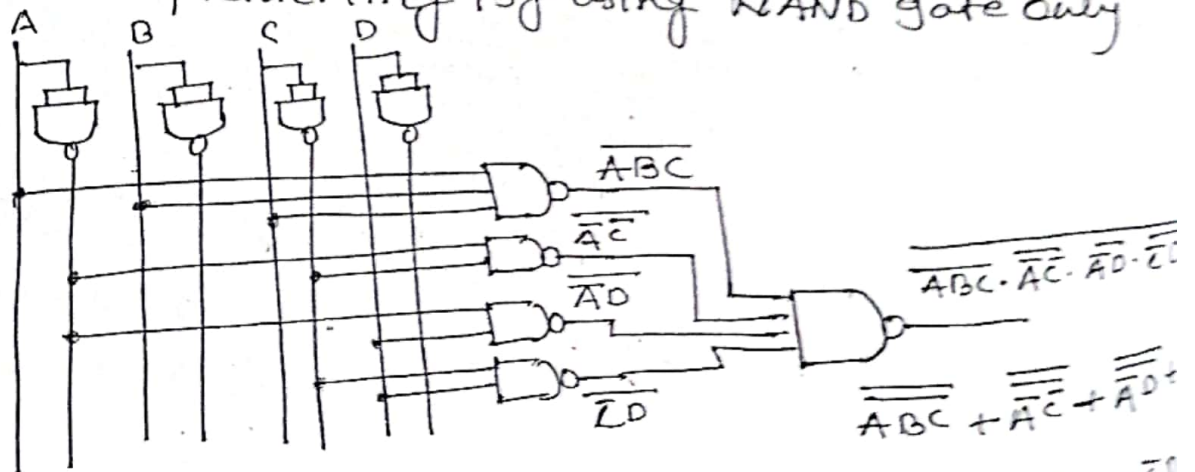
(i) SOP $F = ABC + \bar{A}\bar{C} + \bar{A}D + \bar{C}D$

Now, Double Complementing on Both side we get

$$\overline{\overline{F}} = \overline{\overline{ABC + \bar{A}\bar{C} + \bar{A}D + \bar{C}D}}$$

$$= \overline{ABC \cdot \bar{A}\bar{C} + \bar{A}D + \bar{C}D}$$

Now, Implementing by using NAND gate only



$$F = ABC + \bar{A}\bar{C} + \bar{A}D + \bar{C}D$$

Fig: logic diagram.

Q. A Boolean Function is given by $F(A, B, C, D) = \sum(3, 4, 5, 7, 9, 13, 14, 15)$ and don't care condition $d(A, B, C, D) = \sum(0, 2, 8)$. Simplify it using K-map and implement using NAND gate only. (2014 Fall)

⇒ Here, (A, B, C, D)
 Given $F = \sum(3, 4, 5, 7, 9, 13, 14, 15)$
 $d(A, B, C, D) = \sum(0, 2, 8)$

CD \ AB	00	01	11	10
00			1	
01	1	1	1	
11		1	1	1
10		1		

$$F = \overline{A}B\overline{C} + \overline{A}CD + A\overline{C}D + ABC$$

Now, Double complementing on both side we get

$$\begin{aligned}\overline{\overline{F}} &= \overline{\overline{A}B\overline{C} + \overline{A}CD + A\overline{C}D + ABC} \\ &= \overline{\overline{A}B\overline{C}} \cdot \overline{\overline{A}CD} \cdot \overline{A\overline{C}D} \cdot \overline{ABC}\end{aligned}$$

Now, Implement by using NAND gate only.

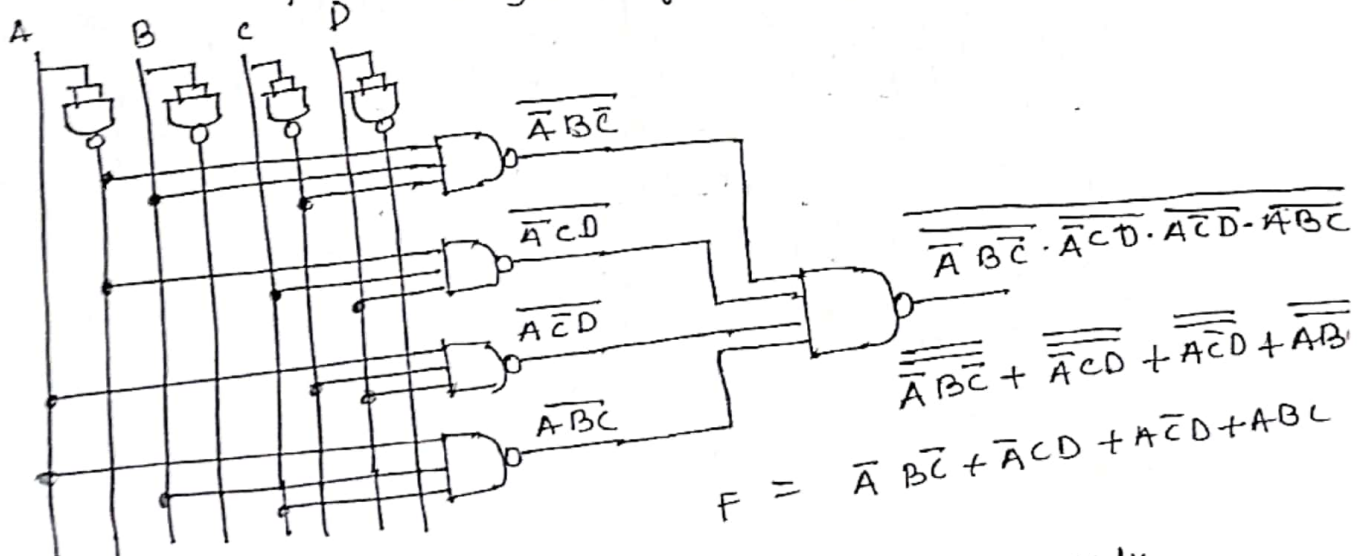


Fig: logic diagram using NAND gate only.

Prepared By: ER. Manoj Kumar Singh

* A Boolean function is given by $F(A, B, C, D) = \sum(3, 4, 6, 8, 10, 12, 14)$ and don't care condition $d(A, B, C, D) = \sum(0, 2, 7)$. Simplify it using K-Map & implement using NAND gate only.

[2015 Fall / Spring]

Solⁿ → Here, Given

$$F(A, B, C, D) = \sum(3, 4, 6, 8, 10, 12, 14)$$

$$d(A, B, C, D) = \sum(0, 2, 7)$$

CD \ AB	00	01	11	10
00	X		1	X
01	1		X	1
11	1			1
10	1			1

$$F = \bar{A}C + \bar{D}$$

Now, Double complementing on both sides we get

$$\overline{\overline{F}} = \overline{\overline{\bar{A}C + \bar{D}}}$$

$$= \overline{\bar{A}C \cdot \bar{D}}$$

Now Implement By using NAND gate only.

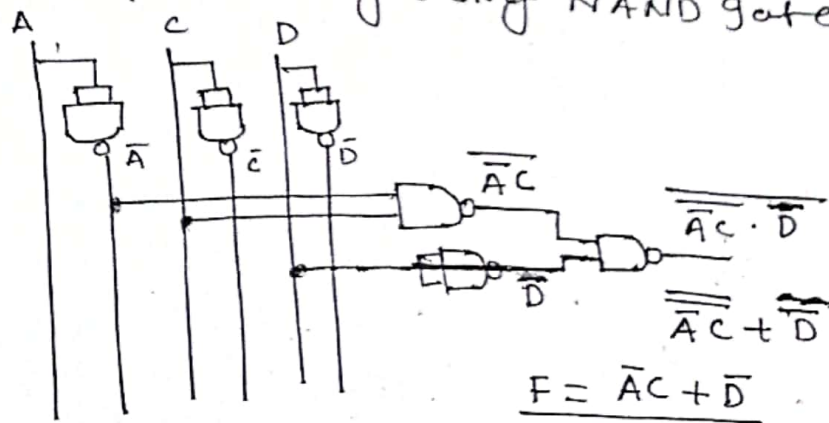


Fig: Logic diagram

Prepared By: GR. Manoj K. S.

Q. Simplify given function using K-Map & implement it in a circuit.

$$F(W, X, Y, Z) = \Sigma(1, 4, 5, 6, 12, 14, 15) \text{ \& don't care}$$

$$\text{Condition } D(W, X, Y, Z) = \Sigma(11, 13)$$

[2016 Spring]

Solⁿ →

$$F(W, X, Y, Z) = \Sigma(1, 4, 5, 6, 12, 14, 15)$$

$$D(W, X, Y, Z) = \Sigma(11, 13)$$

		yz	00	01	11	10
wx	00			1		
	01		1	1		1
	11		1	X	1	1
	10				X	

$$F = \bar{w}\bar{y}z + x\bar{z} + wx$$

Now, Double Complementing on both side we get

$$\begin{aligned}\bar{\bar{F}} &= \overline{\bar{w}\bar{y}z + x\bar{z} + wx} \\ &= \overline{\bar{w}\bar{y}z} \cdot \overline{x\bar{z}} \cdot \overline{wx}\end{aligned}$$

Now, Implement by using NAND gate

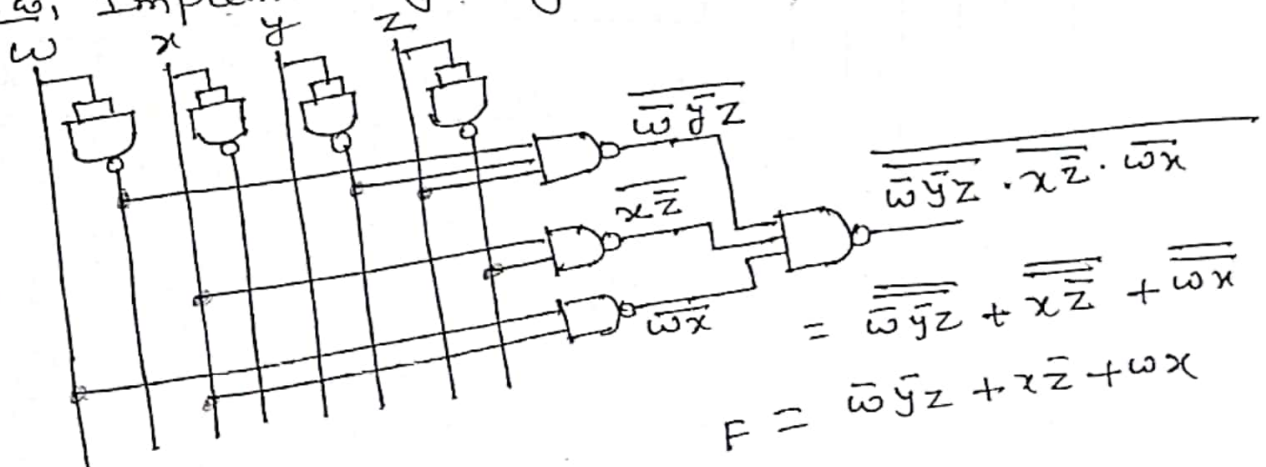


Fig. Logic diagram.

Prepared By: ER. Manoj Kumar Singh

Q. Simplify the following Boolean expression using K-Map & implement using NAND gates only.
 $F(A, B, C, D) = \sum(0, 2, 4, 6, 12, 15)$
 $D(A, B, C, D) = \sum(8, 10, 14)$ [2019 Fall]

Soln

CD \ AB	00	01	11	10
00	1			1
01	1			1
11	1		1	X
10	X			X

$$F = ABC + \bar{D}$$

Now, Double Complementing on both sides we get

$$\begin{aligned}\bar{\bar{F}} &= \bar{\bar{ABC + \bar{D}}} \\ &= \bar{ABC \cdot \bar{D}} \\ &= \bar{ABC} + \bar{\bar{D}} \\ &= \bar{ABC} + D\end{aligned}$$

Now, Implementing by using NAND gate only

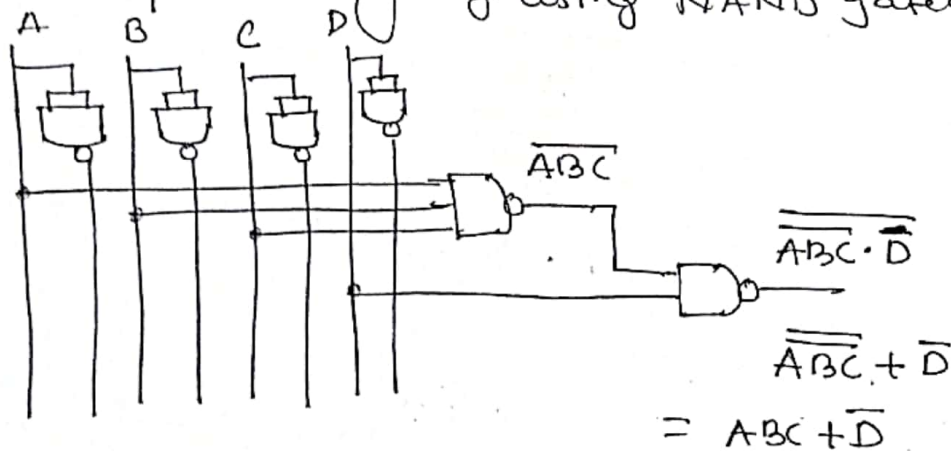


Fig. Logic diagram.

Prepared By: PR. M.

Q. Use K-Map to Simplify the given function & once by considering the don't care condition & once by ignoring the don't care condition & realize it using the basic gates
 $F(A, B, C, D) = \sum(1, 4, 8, 12, 13, 15)$ & don't care $d(A, B, C, D) = \sum(3, 14)$ [2017 Spring]

Soln $\rightarrow$ (i) First case

Considering the don't care condition

$$F(A, B, C, D) = \sum(1, 4, 8, 12, 13, 15)$$

$$d(A, B, C, D) = \sum(3, 14)$$

CD \ AB	00	01	11	10
00		1	X	
01	1			
11	1	1	1	X
10	1			

$$F = \bar{A}\bar{B}D + B\bar{C}\bar{D} + A\bar{C}\bar{D} + AB$$

logic diagram

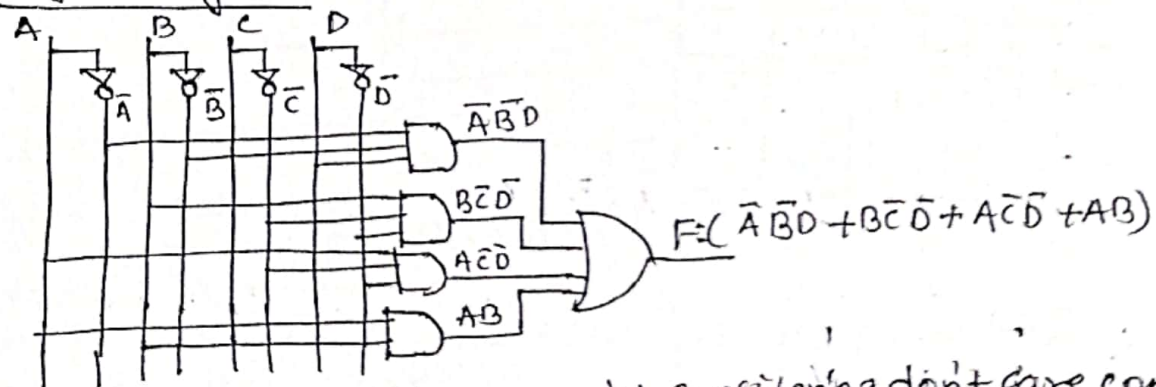


Fig. logic diagram with considering don't care condition

(ii) Second case.

Ignoring the don't care condition.

$$F(A, B, C, D) = \sum(1, 4, 8, 12, 13, 15)$$

CD \ AB	00	01	11	10
00		1		
01	1			
11	1	1	1	
10	1			

$$F = \bar{A}\bar{B}\bar{C}D + B\bar{C}\bar{D} + A\bar{C}\bar{D} + ABD$$

) logic diagram in next page

Prepared By:- E. Manoj Kr. Singh

$$F = \bar{A}\bar{B}\bar{C}D + B\bar{C}\bar{D} + A\bar{C}\bar{D} + ABD$$

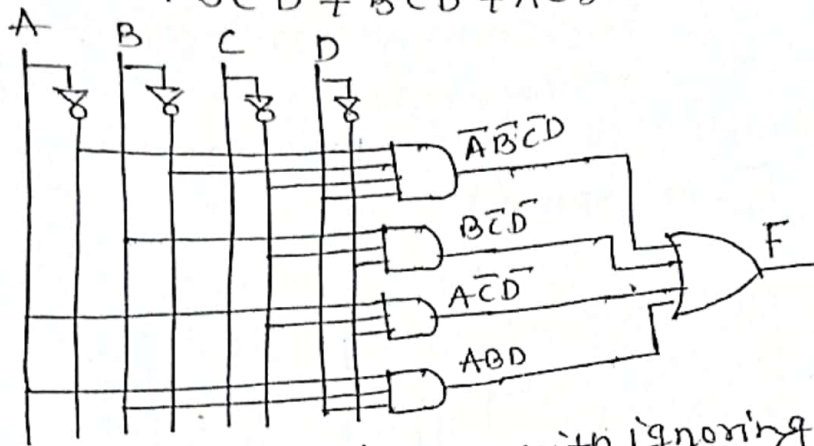


Fig:- logic diagram with ignoring the don't care cond.

Q. Use K-Map to simplify the given Boolean function w/ don't care condition & realize it using basic gates only.
 $F = \sum(1, 4, 8, 12, 13, 15)$ & $d = \sum(3, 7, 11, 14)$ (2018 Fall)
 Soln: Here Given.

$$F = \sum(1, 4, 8, 12, 13, 15) \text{ & } d = \sum(3, 7, 11, 14)$$

CD \ AB	00	01	11	10
00		1	X	
01	1		X	
11	1	1	1	X
10	1		X	

$$F = \bar{A}\bar{B}D + B\bar{C}\bar{D} + A\bar{C}\bar{D} + AB$$

Now, Realizing this expression using Basic gates only.

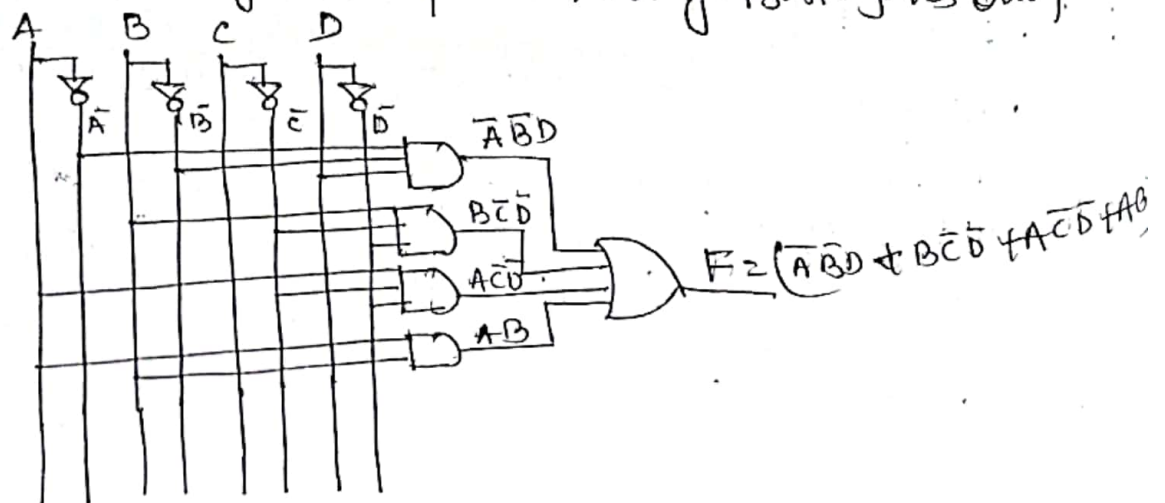


Fig. logic diagram.

Prepared BY: Mr. Manoj Kr. Singh,

* Design a combinational circuit whose input is a four bit number and at the output, we obtain the two's complement of the input.

⇒ Let A_3, A_2, A_1, A_0 be the four bit input & B_3, B_2, B_1, B_0 represents the two's complement of input.

(i) First let us write the Truth table.

Decimal	input				output			
	A_3	A_2	A_1	A_0	B_3	B_2	B_1	B_0
0	0	0	0	0	0	0	0	0
1	0	0	0	1	1	1	1	1
2	0	0	1	0	1	1	1	0
3	0	0	1	1	1	1	0	1
4	0	1	0	0	1	1	0	0
5	0	1	0	1	1	0	1	1
6	0	1	1	0	1	0	1	0
7	0	1	1	1	1	0	0	1
8	1	0	0	0	1	0	0	0
9	1	0	0	1	0	1	1	1
10	1	0	1	0	0	1	1	0
11	1	0	1	1	0	1	0	1
12	1	1	0	0	0	1	0	0
13	1	1	0	1	0	0	1	1
14	1	1	1	0	0	0	1	0
15	1	1	1	1	0	0	0	1

Truth Table.

K-Map Simplification.

For output B_3

$A_3 A_2$	00	01	11	10
00	0	1	4	12
01	1	1	5	13
11	1	1	7	15
10	1	1	6	14

$$B_3 = \bar{A}_3 A_2 + \bar{A}_1 \bar{A}_0 A_3 \bar{A}_2 + A_0 \bar{A}_3$$

$$= \bar{A}_3 A_2 + \bar{A}_1 \bar{A}_0 A_3 \bar{A}_2 + \bar{A}_3 (A_0 + A_1)$$

For output B_2 Group ①

$A_3 A_2$	00	01	11	10
00	0	1	4	12
01	1	1	5	13
11	1	1	7	15
10	1	1	6	14

$$B_2 = A_2 \bar{A}_1 \bar{A}_0 + A_0 \bar{A}_2 + A_1 \bar{A}_2$$

$$= \bar{A}_0 \bar{A}_1 A_2 + \bar{A}_2 (A_0 + A_1)$$

But $\bar{A}_0 \cdot \bar{A}_1 = \overline{A_0 + A_1}$ (De Morgan's)

$$B_2 = A_2 (\bar{A}_0 + \bar{A}_1) + \bar{A}_2 (A_0 + A_1)$$

Let $A_0 + A_1 = X$

$$B_2 = A_2 \bar{X} + \bar{A}_2 X = A_2 \oplus X$$

$$B_2 = A_2 \oplus (A_0 + A_1)$$

For output B_1

$A_3 A_2$	00	01	11	10
$A_1 A_0$	0	1	12	8
01	1	5	13	9
11	3	7	5	11
10	2	4	14	10

$$B_1 = A_0 \bar{A}_1 + \bar{A}_0 A_1$$

$$= A_0 \oplus A_1$$

For output B_0

$A_3 A_2$	00	01	11	10
$A_1 A_0$	0	1	1	1
01	1	1	1	1
11	1	1	1	1
10	1	1	1	1

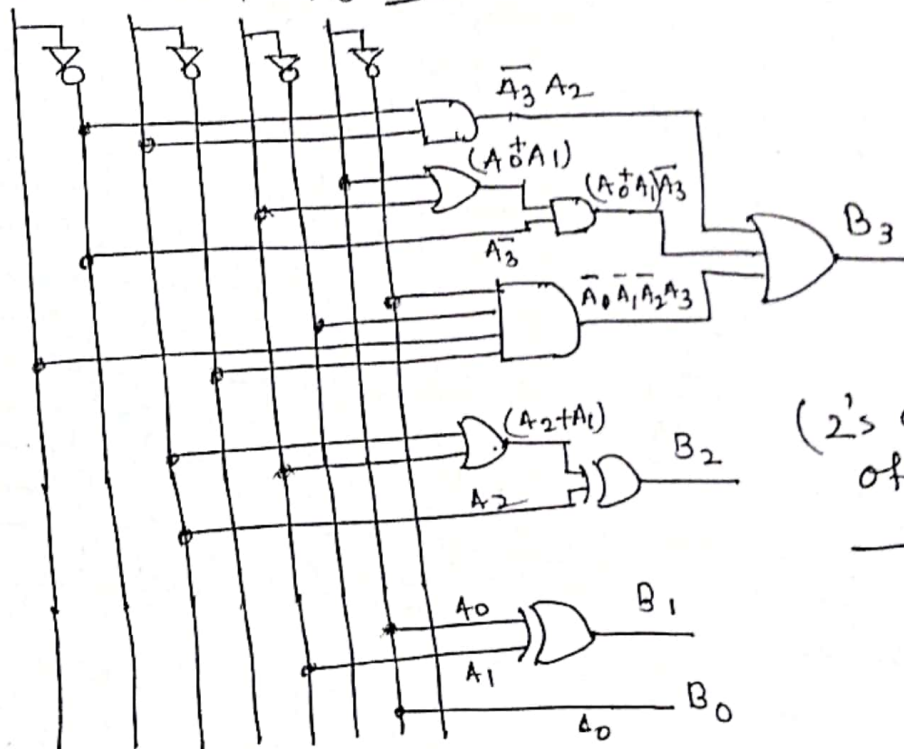
$$B_0 = A_0$$

$$B_3 = \bar{A}_3 A_2 + \bar{A}_1 \bar{A}_0 A_3 \bar{A}_2 + \bar{A}_3 (A_0 + A_1)$$

$$B_2 = A_2 \oplus (A_2 + A_1)$$

Finally let us realize the combinational ckt using gates.

A_3 A_2 A_1 A_0 input



(2's complement of input)

Fig: The Required Combinational Ckt

- Q. Design a code conversion ckt which converts BCD to excess-3 code (2012 fall) (Done in class)
- Q. Design a combinational ckt that has four inputs & two outputs. One of the o/p is high when majority of i/p's are high. Second o/p is high only when all inputs are of same type (2017 spring) (Done in class)

* Design a code conversion circuit which converts Binary code to Gray code. (2019 Fall)

Solⁿ First let us write the truth table

Decimal	Binary code D C B A	Gray code G ₃ G ₂ G ₁ G ₀
0	0 0 0 0	0 0 0 0
1	0 0 0 1	0 0 0 1
2	0 0 1 0	0 0 1 1
3	0 0 1 1	0 0 1 0
4	0 1 0 0	0 1 1 0
5	0 1 0 1	0 1 1 1
6	0 1 1 0	0 1 0 1
7	0 1 1 1	0 1 0 0
8	1 0 0 0	1 0 0 0
9	1 0 0 1	1 1 0 1
10	1 0 1 0	1 1 1 1
11	1 0 1 1	1 1 1 0
12	1 1 0 0	1 0 1 0
13	1 1 0 1	1 0 1 1
14	1 1 1 0	1 0 0 1
15	1 1 1 1	1 0 0 0

Truth table of Binary code to Gray code.

K-Map Simplification:

For G_0

CD \ AB	00	01	11	10
00		1		1
01		1		1
11		1		1
10		1		1

$$G_0 = A\bar{B} + \bar{A}B$$

$$= A \oplus B$$

For G_1

CD \ AB	00	01	11	10
00			1	1
01	1	1		
11	1	1		
10			1	1

$$G_1 = B\bar{C} + \bar{B}C$$

$$= B \oplus C$$

For G_2

CD \ AB	00	01	11	10
00				
01	1	1	1	1
11				
10	1	1	1	1

$$G_2 = C\bar{D} + \bar{C}D$$

$$= C \oplus D$$

For G_3

AB \ CD	00	01	11	10
00				
01				
11	1	1	1	1
10	1	1	1	1

$G_3 = A$

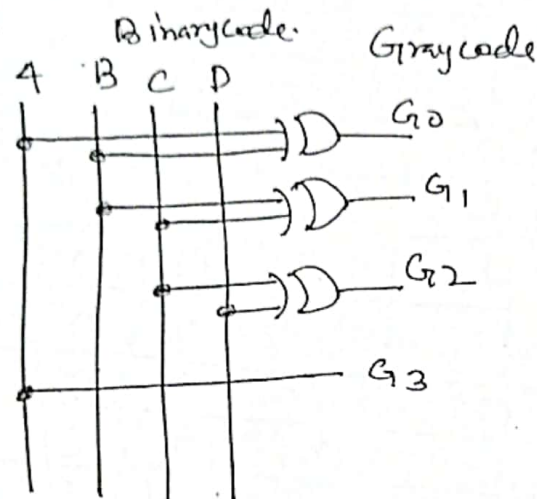


Fig. Logic diagram of Binary to gray code converter.

- * Similarly we can design Gray code to Binary code converter.
- Design a code converter circuit to convert BCD to gray code converter.

BCD code				Gray code			
B_3	B_2	B_1	B_0	G_3	G_2	G_1	G_0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	0
0	1	0	1	0	1	1	1
0	1	1	0	0	1	0	1
0	1	1	1	0	1	0	0
1	0	0	0	1	1	0	0
1	0	0	1	1	1	0	1

Table: Truth Table for BCD to gray code converter

K-Map Simplification

For G_0

$B_3 B_2$	$B_1 B_0$ 00	01	11	10
00		1		1
01		1		1
11	X	X	X	X
10		1	X	X

$$G_0 = \overline{B_1} B_0 + B_1 \overline{B_0} \\ = B_1 \oplus B_0$$

For G_1

$B_3 B_2$	$B_1 B_0$ 00	01	11	10
00			1	1
01	1	1		
11	X	X	X	X
10			X	X

$$G_1 = B_2 \overline{B_1} + \overline{B_2} B_1 \\ = B_2 \oplus B_1$$

For G_2

$B_3 B_2$	$B_1 B_0$ 00	01	11	10
00				
01	1	1	1	1
11	X	X	X	X
10	1	1	X	X

$$G_2 = B_2 + B_3$$

For G_3

$B_3 B_2$	$B_1 B_0$ 00	01	11	10
00				
01				
11	X	X	X	X
10	1	1	X	X

$$G_3 = B_3$$

Logic diagram

Bcd Code.

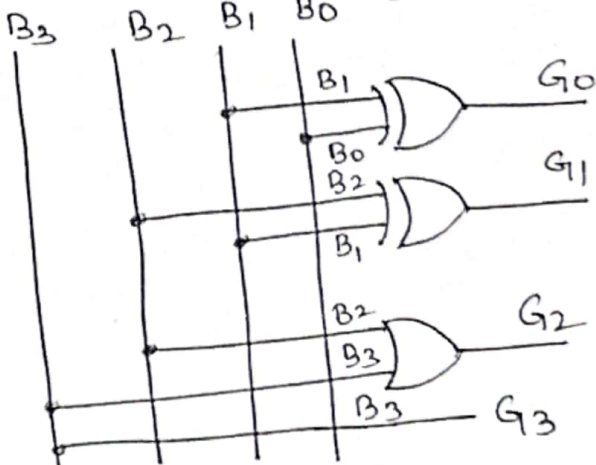


Fig. Bcd to gray code converter

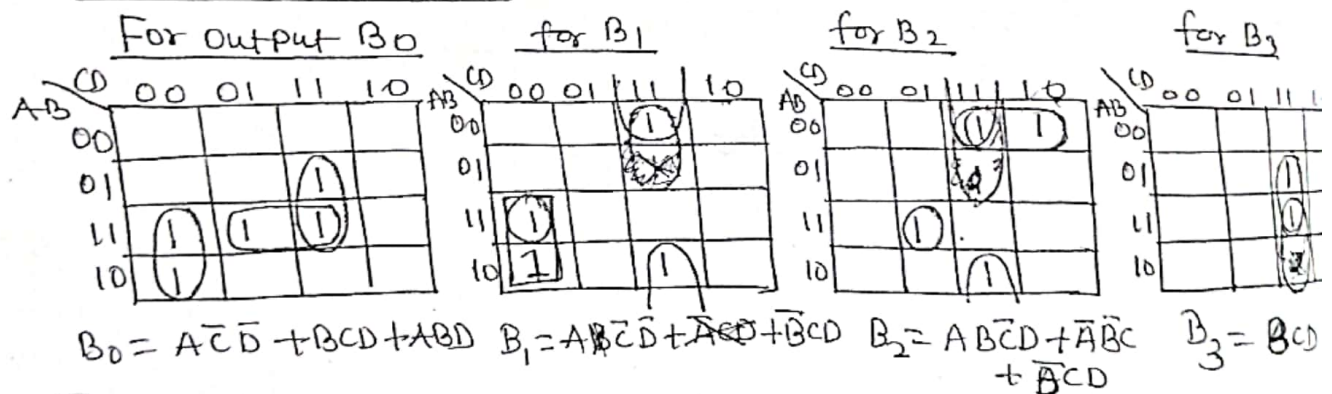
Q. Design a combinational CKT that converts a decimal digit from the 2421 code to BCD (2016 Fall)

⇒ Let us write the truth table

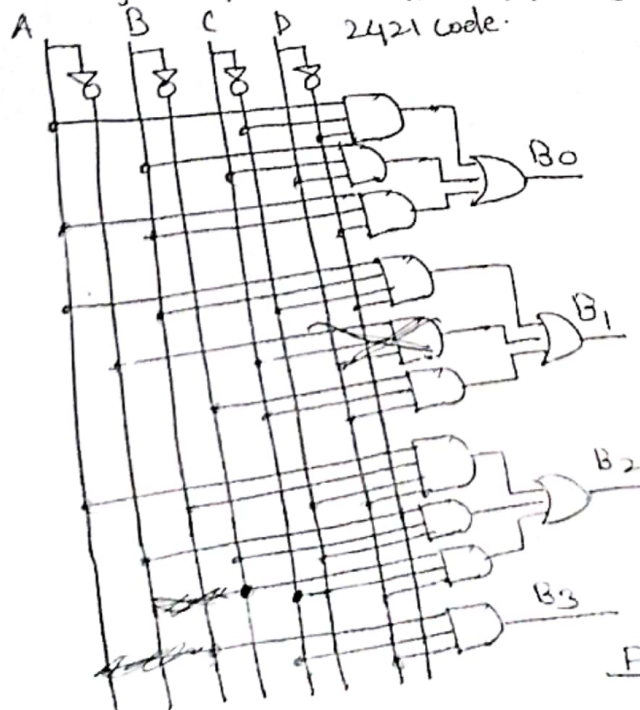
Decimal Digit	2 D	4 C	2 B	1 A	8 B ₃	4 B ₂	2 B ₁	1 B ₀
0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	1
2	0	0	1	0	0	0	1	0
3	0	0	1	1	0	0	1	1
4	0	1	0	0	0	1	0	0
5	1	0	1	1	0	1	0	1
6	1	1	0	0	0	1	1	0
7	1	1	0	1	0	1	1	1
8	1	1	1	0	1	0	0	0
9	1	1	1	1	1	0	0	1

Truth Table of 2421 code to BCD code.

K-Map Simplification:



Finally let us realize the combinational CKT using basic gates



BCD code
8421

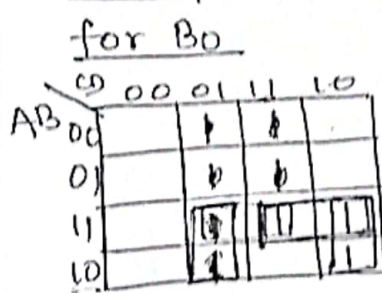
Fig. Required Combinational CKT from 2421 code to BCD

Q. Design a code converter ckt which converts 84-2-1 C to binary. [2012 spring]

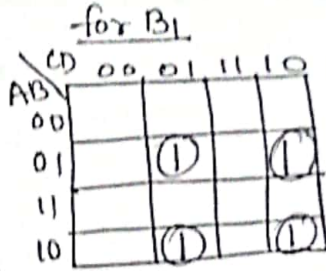
Solⁿ → First let us write the truth table.

Decimal	8 4 -2 -1				Binary			
	D	C	B	A	B ₃	B ₂	B ₁	B ₀
0	0	0	0	0	0	0	0	0
1	0	1	1	1	0	0	0	1
2	0	1	1	0	0	0	1	0
3	0	1	0	1	0	0	1	1
4	0	1	0	0	0	1	0	0
5	1	0	1	1	0	1	0	1
6	1	0	1	0	0	1	1	0
7	1	0	0	1	0	1	1	1
8	1	0	0	0	1	0	0	0
9	1	1	1	1	1	0	0	1

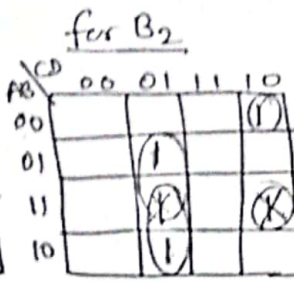
K-Map Simplification.



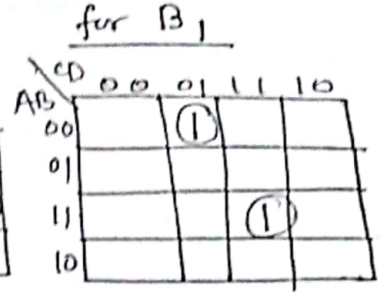
$$B_0 = ABCD + ACBD + ACBD + ACBD$$



$$B_1 = ABCD + ACBD + ACBD + ACBD$$



$$B_2 = ABCD + ACBD + ACBD + ACBD$$



$$B_3 = ABCD + ACBD + ACBD + ACBD$$

Finally. let us realize the combinational ckt using basic gat we set.

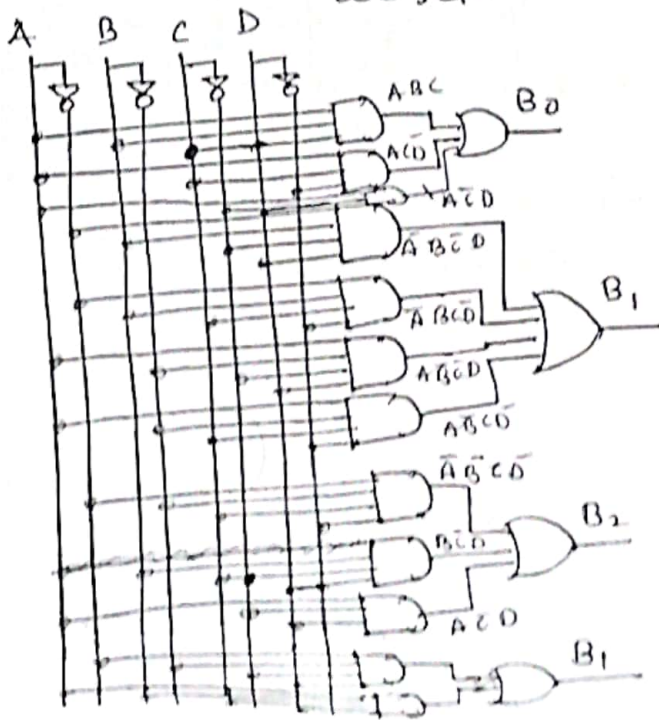


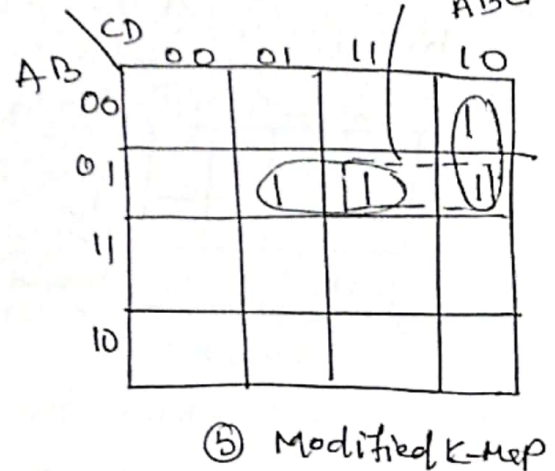
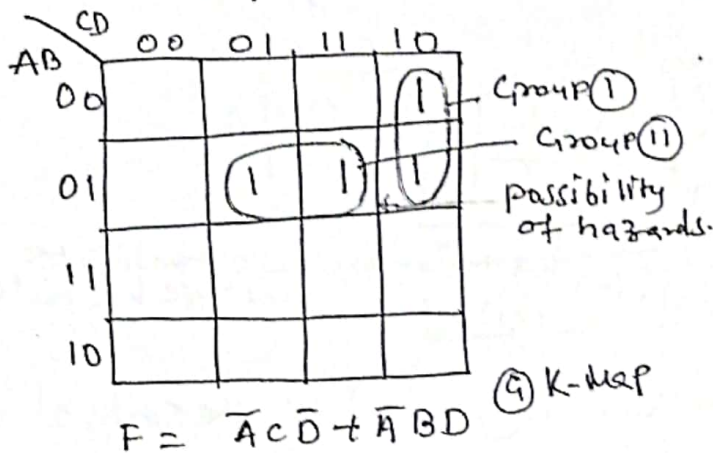
Fig. Required ckt for 84-2-1 code to Binary.

* Design a Combinational Logic Ckt that has 4 inputs numbered 0 through 3, and one Output. The Output is required to go HIGH whenever any one or more, of the input numbers 2, 5, 6, 7, 3, 1, 0. The ckt should be free of static hazards.

solⁿ → First let us write the Truth table.

Inputs				Output Y	
D	C	B	A		
0	0	0	0	0	0
0	0	0	1	X	0
0	0	1	0	0	1
0	0	1	1	0	0
0	1	0	0	X	0
0	1	0	1	X	1
0	1	1	0	X	1
0	1	1	1	0	1
1	0	0	0	0	0
1	0	0	1		0

K-Map simplification for output Y



In fig. the possible point where hazards can occur are shown above. With the normal grouping the expression of o/p will be given by

$$F = \bar{A}C\bar{D} + \bar{A}BD$$

from fig (b) the additional term $\bar{A}BC$ introducing in the output expression. Therefore, expression of the o/p of a hazard free ckt is given by.

$$F = \bar{A}C\bar{D} + \bar{A}BD + \underbrace{\bar{A}BC}_{\text{New term}}$$

The Hazard free ckt is given below

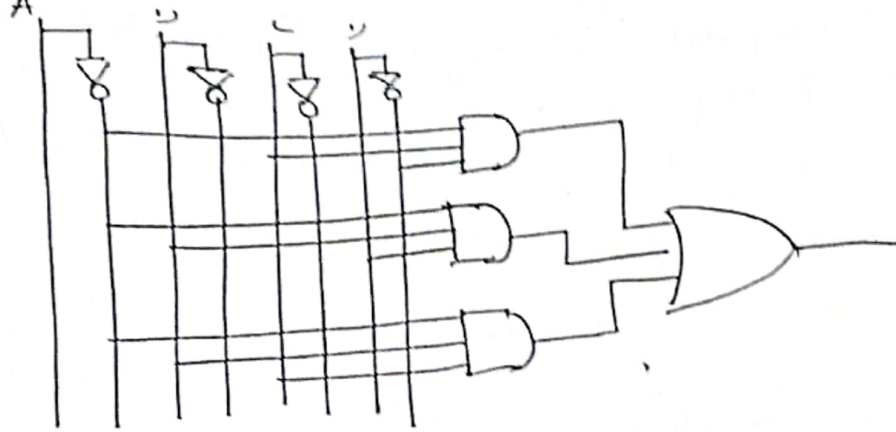
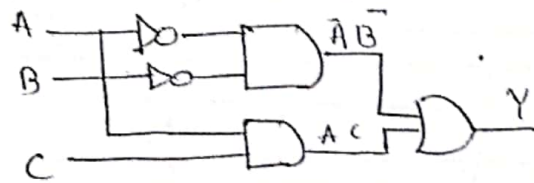


Fig. Realization or Required combinational ckt of Hazard free ckt

* Modify the ckt of fig. to make its hazard free



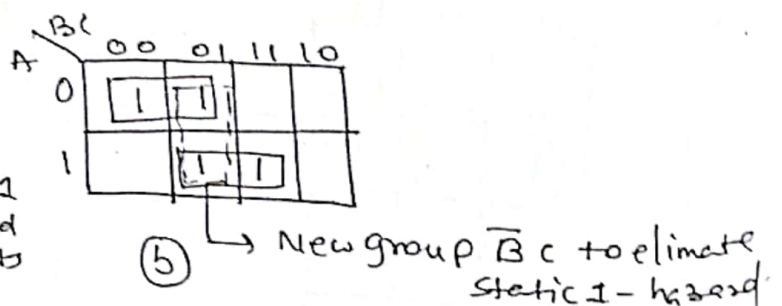
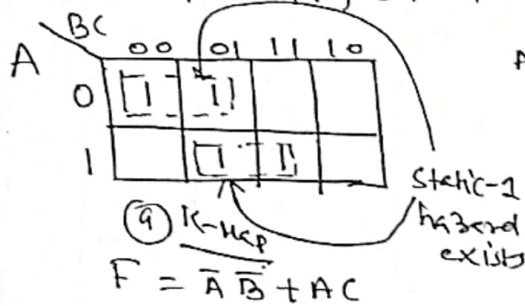
Solⁿ

O/P is Y

$$Y = \bar{A}\bar{B} + AC = \bar{A}\bar{B}(C + \bar{C}) + AC(B + \bar{B})$$

$$= \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C} + ABC + A\bar{B}C$$

K-Map simplification.



The K-Map is Modified as shown in (b) to eliminate static 1-hazard. new expression for o/p will be.

$$F = \bar{A}\bar{B} + AC + \bar{B}C$$

The modified ckt which is free of static 1-hazard.

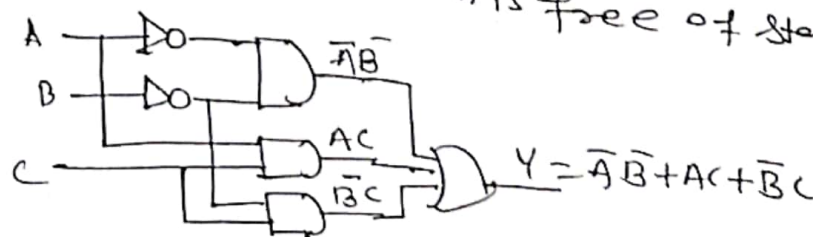


Fig. Hazard free ckt

Q. Implement the following Boolean function $F = \Sigma(1,3,5)$ using 4X1 MUX. (2014 Fall, 2014 Spring, 2015 Fall, Spring, 2016 Spring,

Solⁿ Here, Given that

$$F = \Sigma(1,3,5,6)$$

Let us assume three Variable they are A, B & C.

$$F(A,B,C) = \Sigma(1,3,5,6)$$

Now, Let us write the Truth table of a given function

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Now, Next we write the design table or Implementation table.

	D ₀	D ₁	D ₂	D ₃	
$\bar{A}$	0	①	2	③	Row 1
A	4	⑤	⑥	7	Row 2
Input to MUX	0	1	A	$\bar{A}$	

Fig: Design table

Lastly we draw the logic diagram.

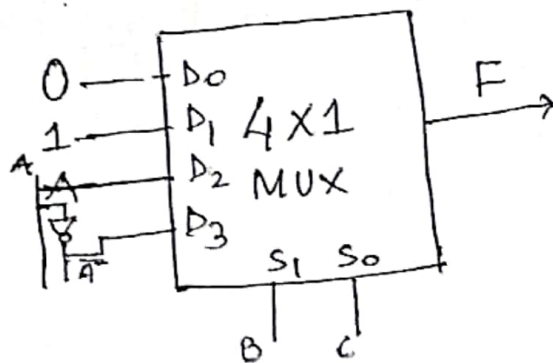


Fig: Logic diagram.

Prepared By: Er. Manoj Kr. Singh.

* Construct 8×1 MUX using 4×1 MUX & explain with truth table. [2019 Fall]

solⁿ → The cascading of two 4×1 MUX results in 8×1 MUX is shown below.

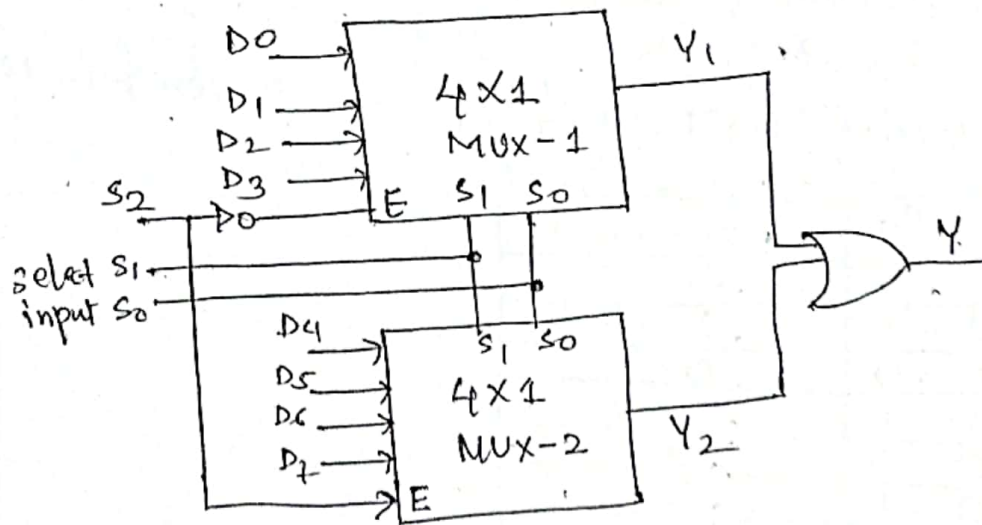


Fig: 8×1 MUX by cascading two 4×1 MUX

Truth Table

select i/p			o/p
s_2	s_1	s_0	
0	0	0	D_0
0	0	1	D_1
0	1	0	D_2
0	1	1	D_3
1	0	0	D_4
1	0	1	D_5
1	1	0	D_6
1	1	1	D_7

MUX1 is enable

MUX2 is enable.

There are in all eight data inputs (D_0 to D_7). The select lines s_1 & s_0 of both 4×1 MUX are connected in parallel whereas a third select i/p s_2 is used for enabling one MUX at a time. The output of the two Multiplexers are ORed to obtain the final o/p Y .

Prepared BY:- Er. Manoj Kr. Singh.

Q. Implement the following logic function using 4x1 MUX
 $F(A, B, C) = \Pi M(0, 1, 3, 5, 7)$

Solⁿ → Here Given Function is

$$F(A, B, C) = \Pi M(0, 1, 3, 5, 7)$$

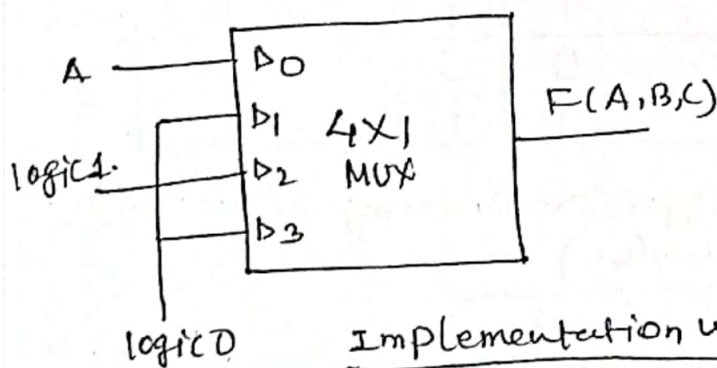
The given expression is in the Pos form so, we must encircle those maxterms which are not included in the given boolean expression.

Implementation table

Inputs	D_0	D_1	D_2	D_3
$\bar{A}$	0	1	(2)	3
A	(4)	5	(6)	7
Input to MUX	$A \quad 0 \quad 1 \quad 0$			

encircle the maxterms which are not included in the given Boolean expression

logic diagram



Q. Implement the following Boolean expression using 8x1 MUX
 $F(A, B, C, D) = \bar{A}\bar{B}\bar{D} + ABC + \bar{B}CD + \bar{A}CD$

Solⁿ → (i) Let us convert the given expression in standard SOP form

$$\bar{A}\bar{B}\bar{D}(C + \bar{C}) + ABC(D + \bar{D}) + \bar{B}CD(A + \bar{A}) + \bar{A}CD(B + \bar{B})$$

$$= \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D} + A\bar{B}C\bar{D} + ABC\bar{D} + A\bar{B}CD + \bar{A}\bar{B}CD + A\bar{B}CD + \bar{A}\bar{B}CD$$

Now, $F(A, B, C, D) = \sum m(2, 0, 15, 14, 11, 3, 7, 3)$

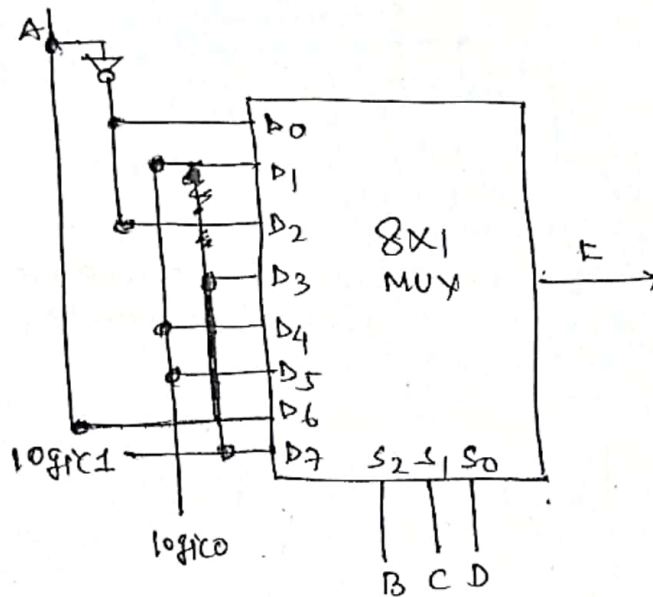
Writing in arranging form

$$F(A, B, C, D) = \sum m(0, 2, 3, 7, 11, 14, 15)$$

Implementation table.

Inputs	D ₀	D ₁	D ₂	D ₃	D ₄	D ₅	D ₆	D ₇
$\bar{A}$	0	1	2	3	4	5	6	7
A	8	9	10	11	12	13	14	15
Input to MUX	$\bar{A}$	0	$\bar{A}$	1	0	0	A	1

(iii) logic diagram.



Implementation

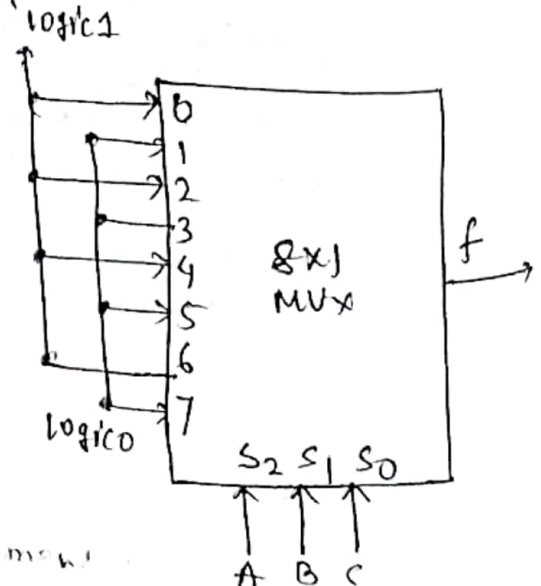
* Implement the following expressing suitable Multiple
 $F(A, B, C) = \sum m(0, 2, 4, 6)$

so, $F(A, B, C) = \sum m(0, 2, 4, 6)$

Here, is three variable, MUX having three select i/p must be
 therefore 8x1 MUX must be used.

Implementation table

D ₀	D ₁	D ₂	D ₃	D ₄	D ₅	D ₆	D ₇
0	1	2	3				



Implementation

Q. Implement the following function using 8-to-1 MUX.
 $Y(A, B, C, D) = \sum(0, 1, 2, 5, 9, 11, 13, 15)$. (2012 Fall, Soln)

Here, Given, $Y(A, B, C, D) = \sum(0, 1, 2, 5, 9, 11, 13, 15)$

First, Let us write the truth table of given function

A	B	C	D	Y
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	1

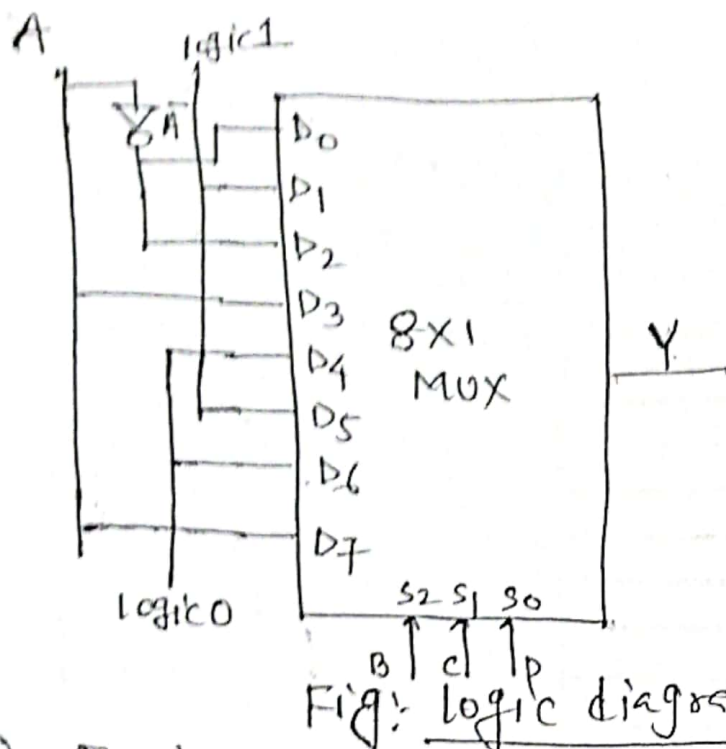
i) Next we write the Implementation table.

	D ₀	D ₁	D ₂	D ₃	D ₄	D ₅	D ₆	D ₇
A'	0	1	2	3	4	5	6	7
A	8	9	10	11	12	13	14	15
Input to MUX	A'	1	A'	A	0	1	0	A

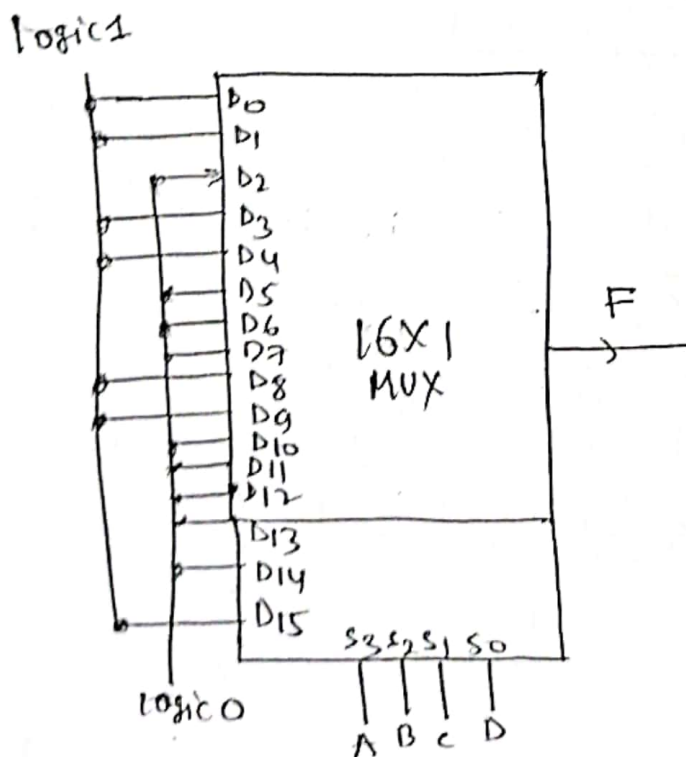
Table: Implementation Table

Lastly we draw the logic diagram

Next Page. PTO



- Q. Implement the following with appropriate MUX.
 (ii) $F(A, B, C, D) = \sum(0, 1, 3, 4, 8, 9, 15)$ (2014 Fall, 2015 Spring)
- Solⁿ Here Given. $(A, B, C, D) = \sum(0, 1, 3, 4, 8, 9, 15)$
 Here is Four Variable. MUX having four select line. So use 16x1 MUX.



Q Implement the Full Subtractor using a 1X8 DEMUX

Solⁿ → The Full subtractor has three inputs A, B and Bin & two O/Ps namely Difference D & borrow out (Bout). The Truth Table of Full Subtractor is

Inputs			Outputs	
A	B	Bin	D	Bout
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

Table:- Truth Table of a F.S

We express the difference & borrow out Outputs in Standard SOP form as

$$D = \sum m(1, 2, 4, 7)$$

$$Bout = \sum m(1, 2, 3, 7)$$

We connect Din to logic 1 permanently & connect A, B & Bin to select S₂, S₁, S₀ respectively.

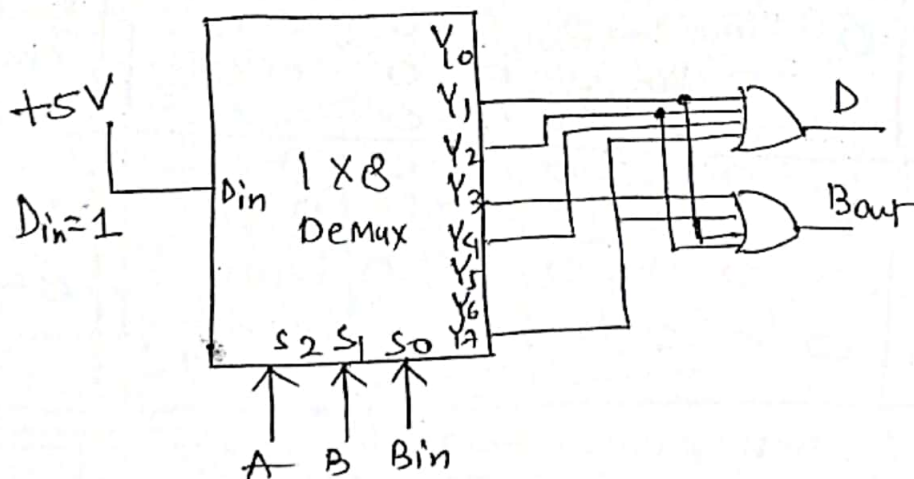


Fig:- F.S using 1X8 DEMUX

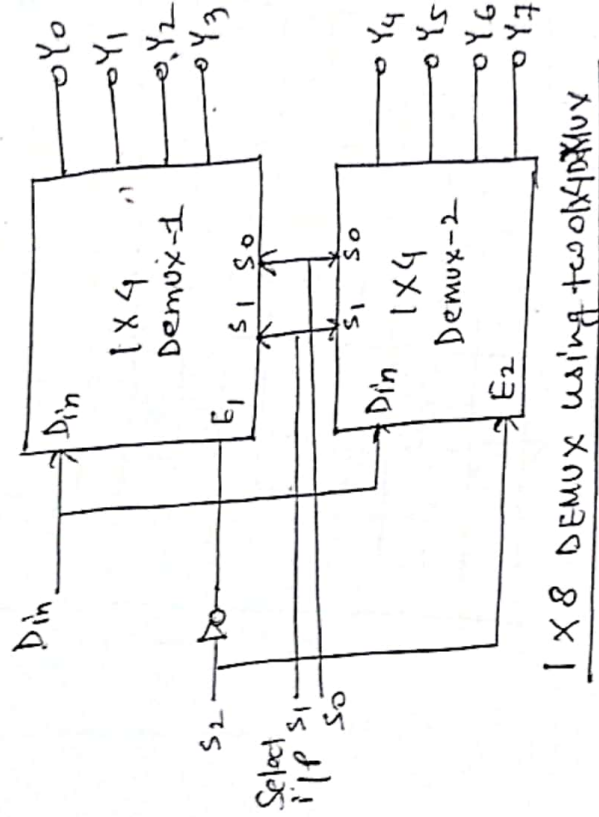
Q. Implement a Full adder using Demultiplexes

Solⁿ → Similarly, we can do as Full subtractor.
(D & Y)

Prepared By:- Er. Manoj Kr. Singh.

* Obtain a 1X8 line DEMUX using two 1X4 line DEMUX.

Soln →



1X8 DEMUX using two 1X4 DEMUX

The select line S_1 & S_0 of the two 1X4 DEMUX are connected parallel with each other & S_2 is used for selecting of two 1X4 DEMUX. S_2 is connected directly to enable (E) i/p of DEMUX-1 whereas inverted S_2 is connected to the enable i/p of DEMUX-2. The T.T of this ckt is shown below.

S_2	S_1	S_0	Y_0	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	Y_7	Demux Enable
0	0	0	Din	0	0	0	0	0	0	0	0
0	0	1	0	Din	0	0	0	0	0	0	0
0	1	0	0	0	Din	0	0	0	0	0	0
0	1	1	0	0	0	Din	0	0	0	0	0
1	0	0	0	0	0	0	Din	0	0	0	0
1	0	1	0	0	0	0	0	Din	0	0	0
1	1	0	0	0	0	0	0	0	Din	0	0
1	1	1	0	0	0	0	0	0	0	Din	0

Table: T.T of 1X8 DEMUX using two 1X4 DEMUX

Q. Implement a Full adder using 4×1 MUX.

solⁿ → (i) First let us write the truth table of Full adder

inputs			outputs	
A	B	C _{in}	Sum(S)	Carry(C)
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

the sum & carry outputs can be expressed in the standard SOP form

$$S = \sum m(1, 2, 4, 7) \text{ \& } C = \sum m(3, 5, 6, 7)$$

ii) Now we write the Implementation table for Sum & carry Output.

For sum

	D ₀	D ₁	D ₂	D ₃
$\bar{A}$	0	①	②	3
A	④	5	6	⑦
Input to MUX	A	$\bar{A}$	$\bar{A}$	A

For carry

	D ₀	D ₁	D ₂	D ₃
$\bar{A}$	0	1	2	③
A	4	⑤	⑥	⑦
Input to MUX	0	A	A	1

Table: Implementation table for Sum & carry.

iii) Logic diagram

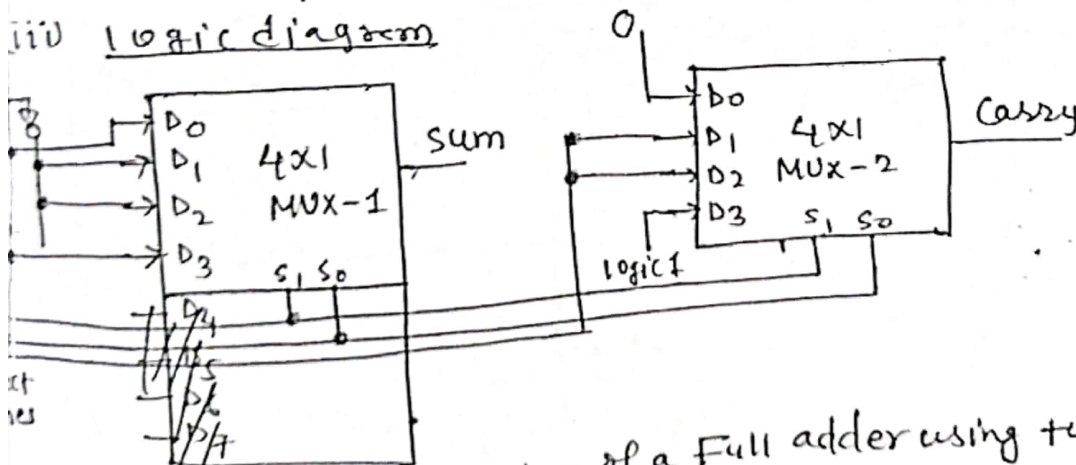


Fig: Implementation of a Full adder using two 4×1 Mux.

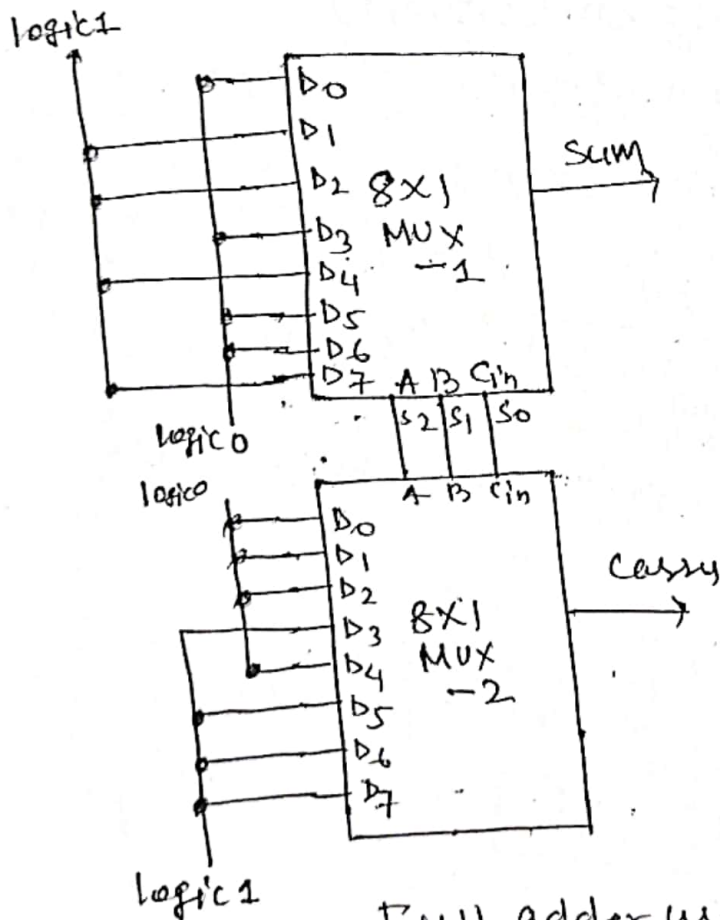
Q. Implement a Full adder using 8X1 MUX.
 Solⁿ (i) First let us write the truth table of a F.A

Inputs			Outputs	
A	B	Cin	S	C
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

(ii) The sum and carry can be expressed in the SOP form
 $S = \sum m(1, 2, 4, 7)$ & $C = \sum m(3, 5, 6, 7)$

(iii) we realize the ckt using ^{two} 8X1 MUX.

Remember
 8 inputs the why
 $2^n = 2^3$
 $n = 3$ sele



Full adder using 8X1 Mux

Note: (we can ^{also} realize ckt separately for sum & carry)

Implement 16x1 multiplexer
 or 1) Implementation is shown below.

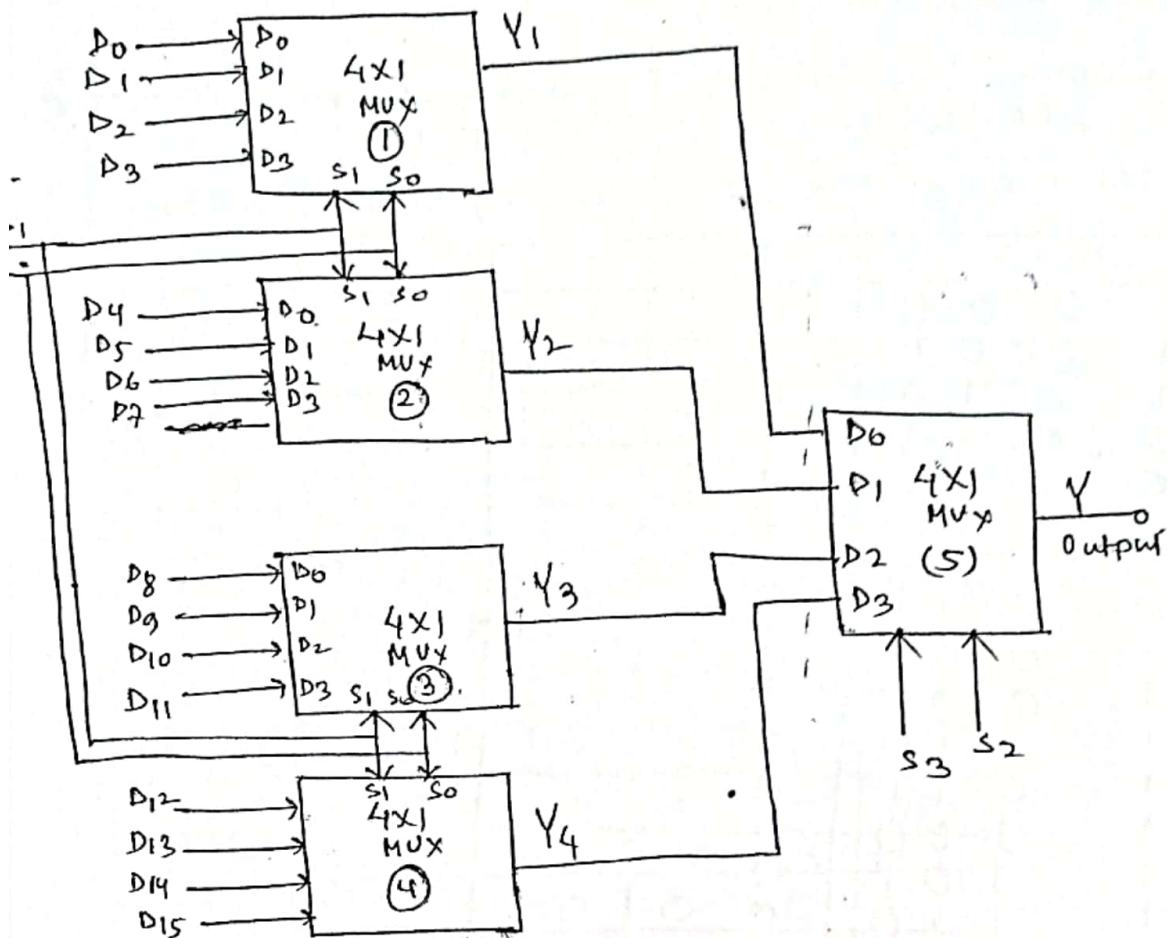


Fig: 16x1 MUX using 4x1 MUX

Function table of above ckt

				MUX O/P				Final O/P Y	
S ₃	S ₂	S ₁	S ₀	Y ₁	Y ₂	Y ₃	Y ₄		
0	0	0	0	D ₀	D ₄	D ₈	D ₁₂	D ₀	S ₃ S ₂ =0 Then for MUX-S select Y
0	0	0	1	D ₁	D ₅	D ₉	D ₁₃	D ₁	
0	0	1	0	D ₂	D ₆	D ₁₀	D ₁₄	D ₂	
0	0	1	1	D ₃	D ₇	D ₁₁	D ₁₅	D ₃	
0	1	0	0	D ₄	D ₅	D ₈	D ₁₂	D ₄	S ₃ S ₂ =01 MUX-S select
0	1	0	1	D ₅	D ₆	D ₉	D ₁₃	D ₅	
0	1	1	0	D ₆	D ₇	D ₁₀	D ₁₄	D ₆	
0	1	1	1	D ₇	D ₈	D ₁₁	D ₁₅	D ₇	
1	0	0	0	D ₈	D ₉	D ₁₂	D ₁₃	D ₈	S ₃ S ₂ =10 Then for MUX- select
1	0	0	1	D ₉	D ₁₀	D ₁₃	D ₁₄	D ₉	
1	0	1	0	D ₁₀	D ₁₁	D ₁₄	D ₁₅	D ₁₀	
1	0	1	1	D ₁₁	D ₁₂	D ₁₅		D ₁₁	
1	1	0	0	D ₁₂	D ₁₃	D ₁₅		D ₁₂	S ₃ S ₂ =11 Then for MUX- select
1	1	0	1	D ₁₃	D ₁₄			D ₁₃	
1	1	1	0	D ₁₄				D ₁₄	
1	1	1	1	D ₁₅				D ₁₅	

The select input S_1, S_0 of MUX 1, 2, 3 & 4 are connected together. The select i/p S_3 & S_2 are applied to MUX-5. The o/p Y_1, Y_2, Y_3, Y_4 are applied to data i/p D_0, D_1, D_2 & D_3 of MUX-5 as shown in fig.

Q. Implement the following function with multiplexer
 sol. $F(A, B, C, D) = \sum(0, 1, 3, 4, 8, 9, 15)$
 First let us write the Truth table of given Function.

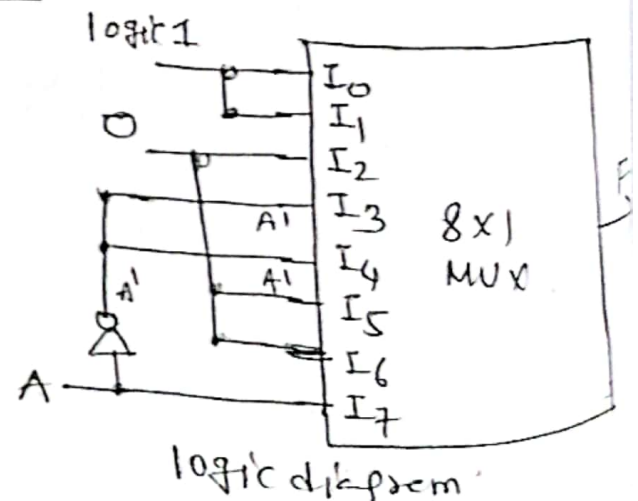
A	B	C	D	F
0	0	0	0	1
0	0	0	1	1
0	0	1	0	0
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

Table-T.T

Implementation Table.

i/p	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
A'	(0)	(1)	2	(3)	(4)	5	6	7
A	(8)	(9)	10	11	12	13	14	(15)
Input to MUX	1	1	0	A'	A'	0	0	A

Finally draw the logic diagram



Q. Design a combinational circuit to produce the 2's complement of a 4-bit binary number.

Soln: $\rightarrow$

Prime NO.	A	B	C	D	P	Q	R	S
0	0	0	0	0	0	0	0	0
1	0	0	0	1	1	1	1	1
2	0	0	1	0	1	1	1	0
3	0	0	1	1	1	1	0	1
4	0	1	0	0	1	1	0	0
5	0	1	0	1	1	0	1	1
6	0	1	1	0	1	0	1	0
7	0	1	1	1	1	0	0	1
8	1	0	0	0	1	0	0	0
9	1	0	0	1	0	1	1	1
10	1	0	1	0	0	1	1	0
11	1	0	1	1	0	1	0	1
12	1	1	0	0	0	1	0	0
13	1	1	0	1	0	0	1	1
14	1	1	1	0	0	0	1	0
15	1	1	1	1	0	0	0	1

K Map for S

AB \ CD	00	01	11	10
00		1	1	
01		1	1	
11		1	1	
10		1	1	

K Map for R

AB \ CD	00	01	11	10
00		1		1
01		1		1
11		1		1
10		1		1

$$R = \bar{C}D + C\bar{D} = C \oplus D$$

S = D K-Map for Q

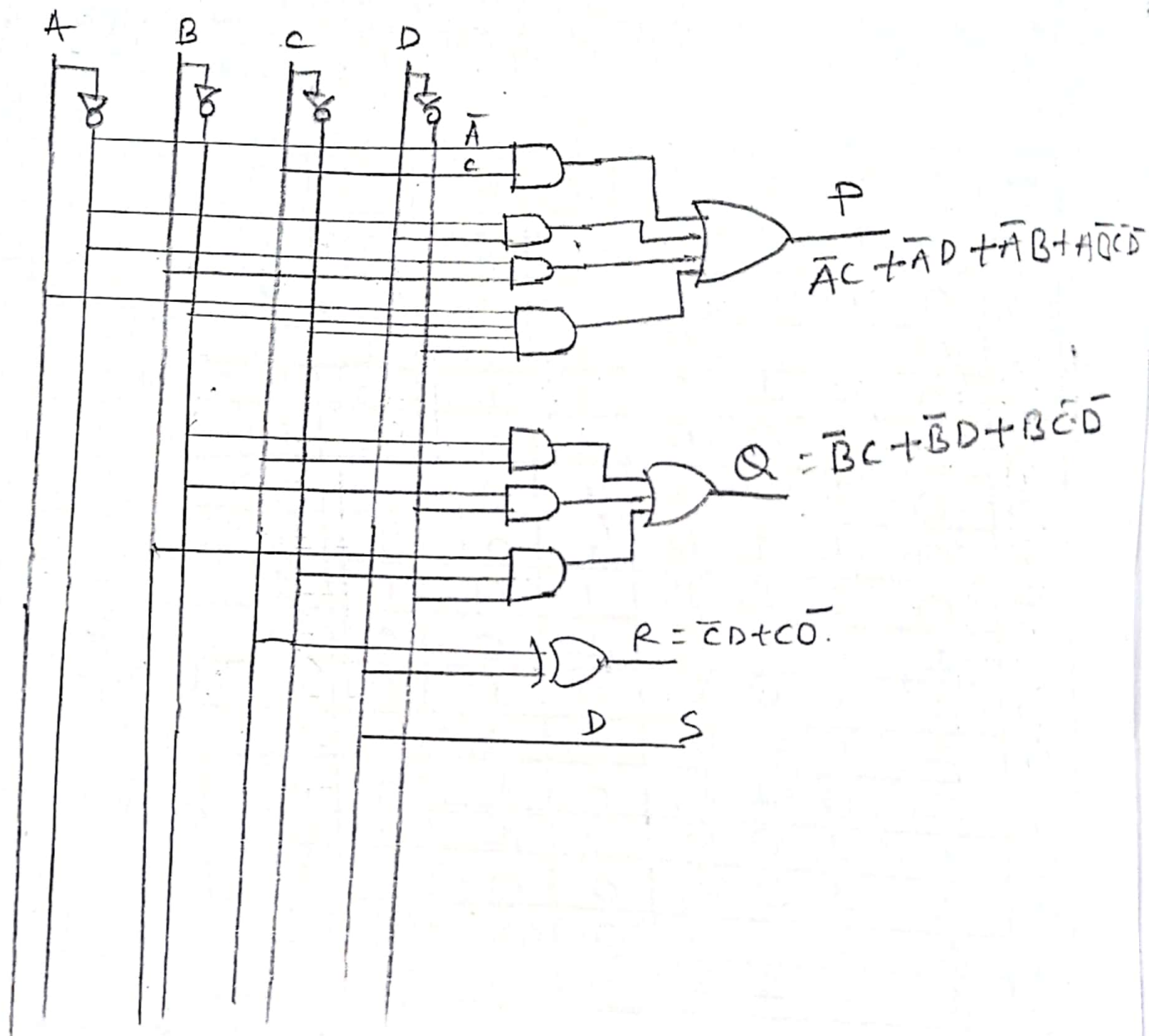
AB \ CD	00	01	11	10
00		1	1	1
01	1			
11	1			
10		1	1	1

$$Q = \bar{B}C + \bar{B}D + B\bar{C}\bar{D}$$

K-Map for P

AB \ CD	00	01	11	10
00		1	1	1
01	1	1	1	1
11				
10	1			

$$P = \bar{A}C + \bar{A}D + \bar{A}B + A\bar{B}\bar{C}\bar{D}$$



Q. Design 3-bit Synchronous UP Counter by using JK F/F

Solⁿ →

$$3 - 8^{1/4} = 3 \text{ F/F}$$

No of state $N = 2^n$

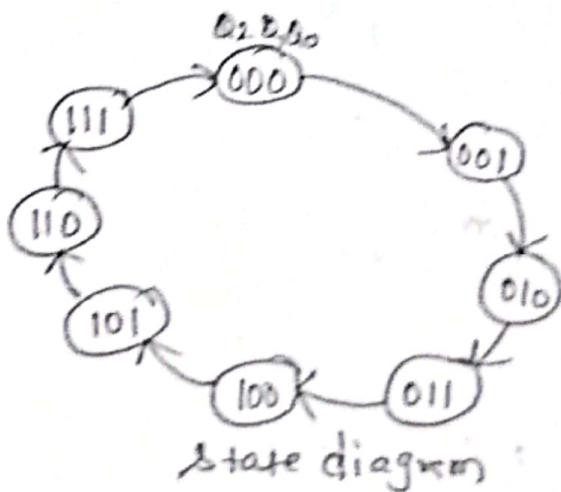
$$n = 3 \quad N = 2^3 = 8 \text{ state}$$

$$\text{Max count } 2^n - 1 = 2^3 - 1 = 8 - 1 = 7$$

Count 0 to 7

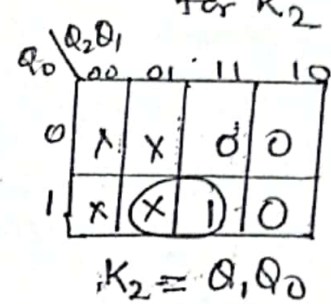
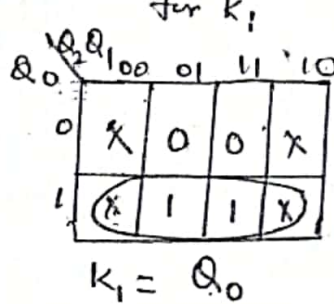
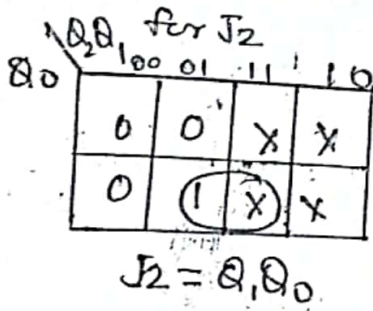
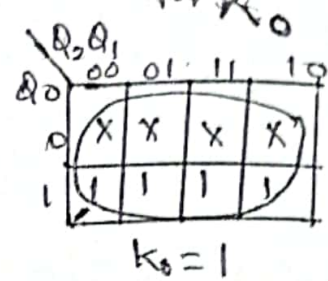
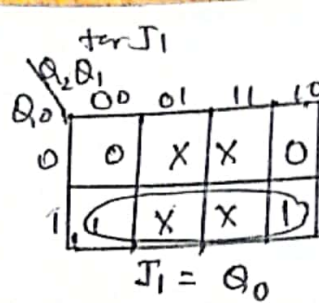
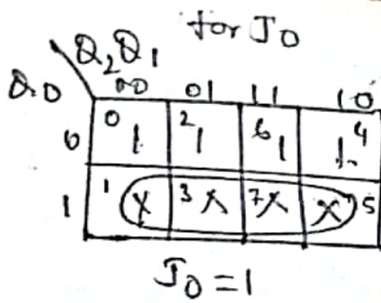
Excitation Table of JK F/F

$Q(t)$	$Q(t+1)$	JK
0	0	0 X
0	1	1 X
1	0	X 1
1	1	X 0

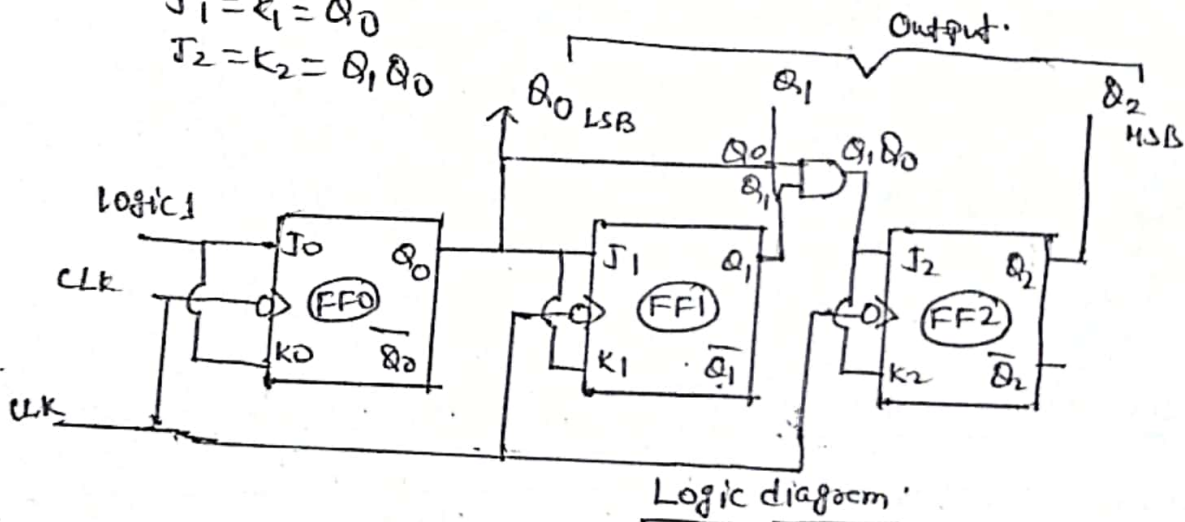


OK+ Excitation Table

Present state $Q_2 \ Q_1 \ Q_0$	Next state $Q_2 \ Q_1 \ Q_0$	$J_0 \ K_0$	$J_1 \ K_1$	$J_2 \ K_2$
0 0 0	0 0 1	1 X	0 X	0 X
0 0 1	0 1 0	X 1	1 X	0 X
0 1 0	0 1 1	1 X	X 0	0 X
0 1 1	1 0 0	X 1	X 1	1 X
1 0 0	1 0 1	1 X	0 X	X 0
1 0 1	1 1 0	X 1	1 X	X 0
1 1 0	1 1 1	1 X	X 0	X 0
1 1 1	0 0 0	X 1	X 1	X 1

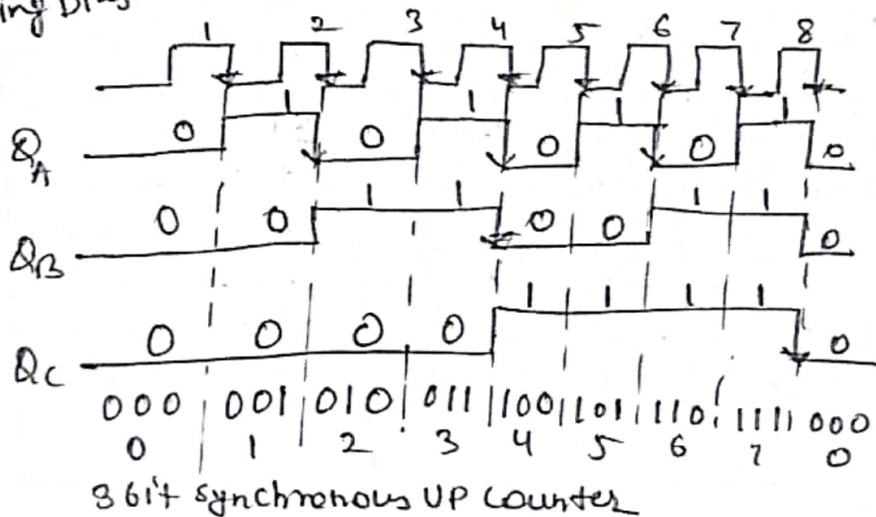


$J_0 = K_0 = 1$
 $J_1 = K_1 = Q_0$
 $J_2 = K_2 = Q_1 Q_0$

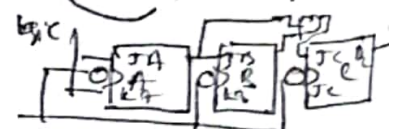


Note:

Timing Diagram



CLK	Q_C	Q_B	Q_A
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1



Here, Q_A changes on each cp as we progress from its original state to its final state & then back to its original state. To produce this operation FF A is held in the toggle mode by connecting $J = K = 1$. For FF B toggle when Q_A is 1. When Q_A is 0 FF B is in no-change mode. Similarly FF C has to change only when $Q_B \& Q_A$ are at 1.

Q. Design 2-bit synchronous UP Counter using JK F/F.

solⁿ →

2-bit = 2 Flip Flop

No. of states $N = 2^n$

$n = 2$ $N = 2^2 = 4$ state

Max. Count $2^n - 1 = 2^2 - 1 = 4 - 1 = 3$

Count = 0 to 3

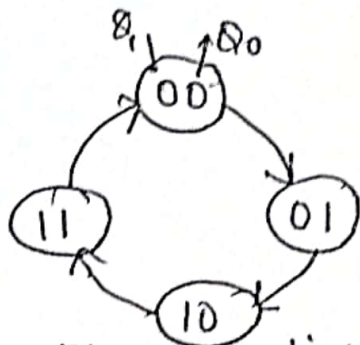


Fig: state diagram

Excitation Table of JK F/F

$Q(t)$	$Q(t+1)$	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

ckt Excitation Table.

PS		NS		I/P			
Q_1	Q_0	Q_1	Q_0	J_1	K_1	J_0	K_0
0	0	0	1	0	X	1	X
0	1	1	0	1	X	X	1
1	0	1	1	X	0	1	X
1	1	0	0	X	1	X	1

for J_1

Q_0	0	1
0	0	1
1	X	X

$J_1 = Q_0$

for K_1

Q_0	0	1
0	X	X
1	0	1

$K_1 = Q_0$

for J_0

Q_1	0	1
0	1	X
1	1	1

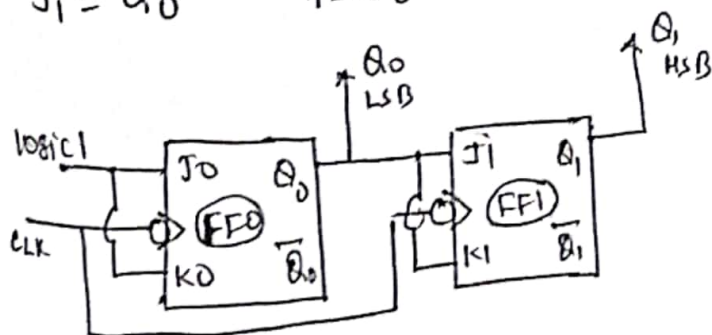
$J_0 = 1$

for K_0

Q_1	0	1
0	X	1
1	X	1

$K_0 = 1$

$J_0 = K_0 = 1$
 $J_1 = K_1 = Q_0$



Logic diagram